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RESEARCH ARTICLE

A New Multi Objective Video Summarization Approach for Video Surveillance Analytics Applications on Smart Cities

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ABSTRACT Summarizing surveillance videos used in smart city applications is very important in terms of transaction costs and sustainability. Although there is a considerable amount of literature on video summarization, the methods in the literature for summarizing surveillance videos used in smart cities are few and inadequate. Because, both object and event features of the mentioned video data must be preserved. In this study, we integrated an object-centric and an event-centric summarization method with Apache Kafka for effective summarization of such videos. With the object-centric summarization module of our proposed method, we focused on preserving the statistical and motion features of the objects in the videos. With the event-centric summarization module, we ensured the preservation of abnormal events in the videos. We presented the performance results of both modules and the integrated system in detail with different metrics. Finally, we compared the performances of both modules with the video summarization approaches in the literature based on different metrics. The developed object-centric summarization method preserves the statistical features of the video with a success rate of over 90% and shortens the videos with a rate of over 95%. The developed event-centric summarization approach provides summaries that include abnormal situations found in videos with a success rate of over 95%. The presented comparative results prove that this original method we developed is superior to the studies in the literature in terms of performance and many different evaluations.

INDEX TERMS Video summarization, video classification, video analytics, transformer, smart city.

I. INTRODUCTION

Smart cities are called high-tech cities that utilize the Internet of Things (IoT), big data and artificial intelligence (AI) technologies in the components of a city such as education, health, security, energy and transportation. In smart city applications performed on the above components of a city, analysis of data obtained from different sensors is utilized. The most commonly used sensor data in applications developed within the scope of smart cities is videos obtained from

imaging devices. Therefore, video analytics approaches find a wide place in smart city applications. In this study, it is aimed to develop a new video summarization approach that summarizes video data obtained within the scope of smart city applications with a multi-purpose approach. In this way, it is aimed to reduce the storage and computational costs caused by smart city applications using video analytics and to make such applications more sustainable with less server usage. In order to implement the proposed method, video analytics applications developed within the scope of the smart city concept in the literature were first examined, and these studies will be explained in detail in the following sections

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(Video Analytics Applications on Smart Cities). Additionally, different approaches focused on video summarization in the literature have also been examined and presented under this study. In this section of the study, approaches focusing on video summarization will be included.

In a study, video summarization approaches using machine learning were discussed to eliminate the memory cost and user time loss caused by the increase in video data with the developing technology [1]. In the relevant study, video summarization approaches were categorized according to being single and multiple views. Again, one of the main points emphasized by the study is that the effectiveness of the summarization approaches increases by determining the features used in summarization with deep learning-based methods. The study focuses on current approaches, data sets, applications, problems, and future field opportunities related to video summarization. Before giving a detailed explanation of these applications, some concepts frequently used in video summarization applications are explained [1]. It would be very useful to include these concepts in our study. The concept of video skim refers to the important video parts obtained for the production of a summary video from a video. A Storyboard is a visual representation of the video, scene by scene. Keyframes refer to the most important frames obtained from the original videos. Video streaming refers to the video transmission process that continues from the source to the monitor. In addition, video summarization approaches are considered in two categories as static video summarization and dynamic video summarization [1]. Static summaries are video summaries created by grouping important image frames of the original video. In such approaches, key frames determined on the original video are added to the summary video in chronological order. In dynamic summarization approaches, key scenes and chunks of the original video are included in the summary. Skims consist of segments and audio sections consisting of multiple frames of the video; that is, visual and audio features of the video stream are utilized. In dynamic summarization approaches, the target object, event, perception, and feature-based video summarization can be performed. In the light of the information obtained from the above study, it is understood that our approach has dynamic video summarization features.

Another study focused on examining deep learning-based video summarization approaches in the literature [2]. In this context, video summarization approaches were first classified in terms of modality in the relevant study. While unimodal approaches perform video summarization only by focusing on feature extraction and summarization learning, multimodal approaches also benefit from textual metadata and semantic relations of the video category [2]. Again, approaches in the literature are categorized as supervised, unsupervised and weakly-supervised according to the learning method [2]. Supervised approaches utilize datasets whose ground truth annotations are determined by humans. Unsupervised approaches are approaches that consist of only

quantitatively sufficient data for training and do not need any ground truth processing. Since validation is not performed with a manual ground truthing process in such approaches, the time loss in the development of the method is considerably less compared to the first approach. In weakly-supervised approaches, ground truth labeling is simply performed. The aim of using approaches other than supervised approaches in summarization applications is to prevent inefficiencies caused by human error.

In another study, a static video summarization method was developed with a multi-objective optimization approach [3]. In the proposed method, color, texture and shape features were used for the summarization process. Summary videos were created with a metaheuristic optimization approach called clustering and artificial moss algorithm. In a study, keyshot selection process was performed for video summarization with a neural architecture of encoder-decoder nature [4]. In the study, a new summarization architecture was developed using CNN (Convolutional Neural Network), RNN (Recurrent Neural Network) and multiple neural layer groups with attention mechanism. The evaluation of the related approach was carried out on two data sets and 44% and 60% successes were obtained. In a study, a feature-constrained video summarization approach was developed [5]. In the related study, firstly, candidate image frames representing the entire video were determined. Then, the importance of these candidate image frames was determined according to their features such as color, object, difference and tracking. At this point, the researchers performed the summarization process by using the regret minimization technique frequently used by database communities. The proposed method has been evaluated on different data sets. As a result of the evaluations, it has been observed that the summarization performance of the proposed method achieves 60% F-score. In another study, a multimodal neural network approach has been used to summarize cricket sports videos [6]. The neural network pipeline used in the study uses visual and audio features. In the first stage, cricket bowling activities are analyzed, and the obtained segments are sent to the second network. The importance of the video piece is determined by the audio analysis performed here. According to the authors, creating dynamic summaries of sports videos by preserving the domain knowledge, content and exciting features with multimodal approaches is more effective since audio and visual features are used together. In another study, behavioral profiles have been used to perform an adaptive video summarization process [7]. In the proposed method, different artificial intelligence and computer vision approaches have been used to perform personalized video summarization. An approach has been proposed that summarizes the sections where people are detected in the video to be used for suspicious situation detection applications in another study [8]. The performance of the module that performs the human detection process is 99.97% in the study. A new multimodal approach was proposed for performing

video summarization within the scope of video subject [9] in an proposed approach. In the approach mentioned, movies on different subjects are labeled and important parts of these videos are determined. Then, these features are taught to a multimodal transformer. As a result of the study, the summarization process was performed with 45.89% success. In another study, a new approach was proposed for summarizing event-centered videos [10]. In the study, event boundaries were determined within the video and a signal expressing event importance was created. This obtained signal was used for the summarization process. In another study, a k-means and quick-sort based summarization method was proposed for video surveillance systems [11]. In the related study, spider monkey optimization is used for the selection of features used in summarization. Then, the normalized selected features are clustered with the k-means method. The image clustered with k-means is included in the summary video with distance-dependent approaches.

In data summarization process, there are distinctions between extractive and abstractive summarization approaches, especially in text summarization approaches [12], [13]. The sentences in the original text are used in the summary text as they are in the extractive summarization. In the abstractive summarization process, the sentences in the summary text are created in an original way by using a neural architecture. In this context, some video summarization approaches in the literature have also been given abstractive qualities [14], [15], [16]. Neural architectures were used to obtain natural language descriptions of videos in one of them [14]. Studies focused on summary text production are evaluated with ROUGE metrics. The ROUGE-1, ROUGE-2 and ROUGE-L metric evaluations obtained in the study are 40.21, 17.64 and 36.89, respectively.

As mentioned above, video summarization approaches can be performed feature or shot-based. In a study, a comprehensive key frame selection method that can be used for both types of summarization processes was proposed [17]. In the proposed method, not only the features obtained from static images, but also the motion information in the video were used. Attention mechanisms were also used in the neural architecture that performs the summarization process. Although the developed neural architecture uses input data with fewer shots per frame than similar shot-based summarization approaches in the literature, when the comparative results are examined, it is seen that it achieves a performance close to other literature studies.

When the studies focused on video summarization in the literature are examined Video summarization approaches have differences such as;

- Dynamic or static approach to video summarization
- Multi-modal or single modal,
- Being feature or event-centered,
- Multi-view or single-view,
- Extractive or abstractive.

Figure 1 provides a diagram that classifies video summarization approaches in the literature according to these

features. When the existing smart city video summarization approaches in the literature and the video analytics applications we provide in the next section are examined, it is seen that these studies do not preserve important information in terms of smart city applications in summary videos in a wide scope. As you will see in the examples we provide in the next section, the focus in SC video applications is on event and statistics-based features. However, existing studies generally address one of these focal points. In addition, object-centric studies have focused on adding video frames containing only a certain object type to the summary video. Summary videos do not preserve general statistical information such as motion and density of objects. In this study, we aim to eliminate these deficiencies of existing applications in the literature.

II. VIDEO ANALYTICS APPLICATIONS ON SMART CITIES

There are many studies in the literature focusing on smart city applications of video analytics. A systematic review of video analytics approaches of this nature operating on the edge was conducted [18] in one of these. As it is known, in smart cities and different IoT systems, sensor data is obtained via the edge, and some analyses that can be performed on the data can be performed on the edge side, thus reducing the processing and computational costs on the cloud. The basic benefits and video analytics components provided by video analytics performed on the edge side are explained in detail in the mentioned study. The benefits determined within the scope of the study are expressed as automatic and effective video observation, incident detection and prevention, and providing a business intelligence platform in this context. In addition, the components of the relevant video analytics applications are expressed as video inputs, video management systems, object detection and tracking, incident detection, data storage, reporting, and user interface. Video input sources in smart cities are surveillance cameras, traffic cameras, and social network feeds. In video management systems, stream processing, video indexing, and storage processes are managed. Machine vision applications developed within the scope of smart cities were examined in a study [19]. A wide coverage is given to video summarization approaches in the video analysis applications discussed here. Just like the video summarization studies we explained in the first section, the video summarization methods focused on in this study are classified according to two different approaches.

These are video summarization methods based on key-frame detection and video summarization methods based on video skimming. In addition to these, the characteristics such as the dynamic and static nature of the video summarization process and the output summary being in image or text format are presented as distinguishing factors for the classification of summarization methods. In the study, sample studies are given for the activities of video summarization approaches in application areas such as sports, movies, security, and academics. Among these, the video summarization area that is most compatible with the smart city concept is surveillance

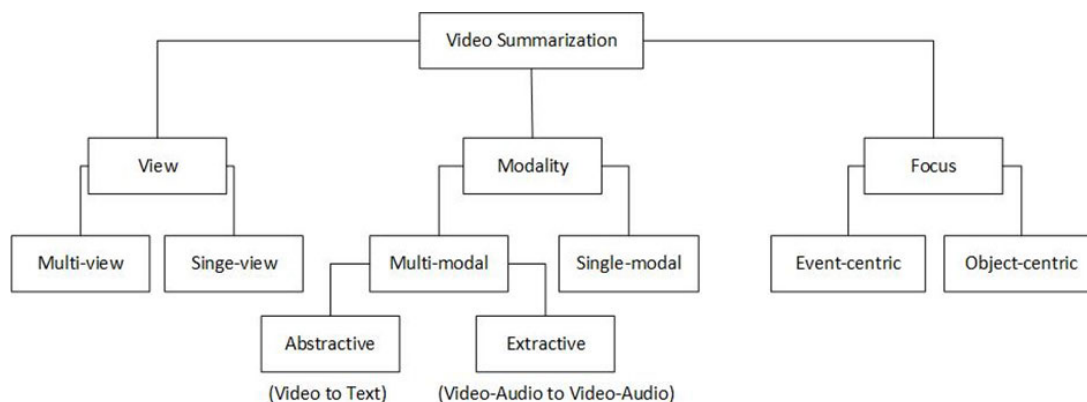


FIGURE 1. Video summarization characteristic tree.

systems. In the study, it is emphasized that the summarization applications made for video surveillance systems are generally carried out with event detection-based approaches. In the relevant literature study, a 5-layer framework is proposed for machine vision applications that will operate within the scope of smart cities. The first layer of the proposed framework includes smart city applications focused on video analysis. The second layer contains video sources. The third layer includes frame enhancement and unnecessary frame removal steps to be applied within the scope of video preprocessing. The fourth layer includes manual and automatic feature extraction steps. The last layer includes video summarization, classification, recognition and detection processes. Finally, in the study, video analysis applications in smart cities were discussed within the scope of different smart city components [19]. These are;

- Video analysis in surveillance applications: It refers to applications performed on videos obtained from surveillance devices in smart homes, buildings, and public areas. Within the scope of this field, video analysis applications such as anomaly detection, traffic monitoring, crime, accident, suspicious event detection, and facial recognition are performed.
- Entertainment analysis in sports and entertainment activities: In sports and entertainment videos, applications such as audience counting, crowd behavior detection, player movement detection and tracking, commentator speech analysis, highlight determination and summarization are usually performed. Multi-modal feature extraction techniques can also be used in such applications.
- Video analysis within the scope of smart health applications: online consultation, smart ambulances, robot assistants and disabled care applications can be given as examples.
- Video analysis within the scope of smart education applications: Examples include producing summaries of educational videos and determining the attention status of students in class.

- Video analysis within the scope of smart vehicles: Examples include video-based obstacle detection and real-time behavior analysis of drivers.
- Video analysis in smart traffic control applications: Examples include smart imaging, traffic density analysis, automatic license plate reading, smart parking, and detection of abnormal activities on the roads.
- Video analysis within the scope of smart forensic applications: Monitoring and detection of criminal activities, determining the originality of video recordings.

Smart cities can be associated with all digital components of a city. Because it is a very wide application area in terms of scope. Therefore, in the mentioned study, video summarization approaches related to entertainment components such as sports, movies and educational components are also included in the smart city concept. The target smart city component that will use the summaries provided by the video summarization approach we developed is different smart video analytics applications rather than humans. In addition, it is aimed to obtain evidence that will verify the results of existing smart city applications applied on video by obtaining these summaries. As can be seen in the studies we will give below, video analytics applications in smart cities are mostly carried out on videos taken from surveillance systems. The motivation, objectives and focus of the developed study will be explained in detail in the next section.

In another study, the methods, data sets, evaluation approaches and hardware requirements used for video analytics applications in smart cities are explained in detail [20]. In the related study, the neural architectures such as MobileNet, SingleShotDetector, YOLO (You Look Only Once), EfficientDet were explained in detail and the performance comparisons of these models were performed for different scenarios. The study focused on approaches that perform video analytics applications such as object detection and tracking on edge resources. In addition to the performance comparison of the models, performance comparisons were also performed for video analytics applications performed

with these models on edge IoT devices such as Raspberry Pi and Google Coral.

A comprehensive review of artificial intelligence (AI) supported visual IoT applications in the literature was conducted [21] in a study. The applications examined were classified as smart surveillance, smart transportation, weather monitoring, sports, health, security and public safety according to their fields in the mentioned study. Within the scope of smart surveillance applications, human, vehicle and other moving object detection applications, video or image enhancement applications, image background and foreground detection applications, human behavior detection, video segmentation applications, simple video analytics approaches executed on edge devices were also examined. Within the scope of smart transportation applications, applications such as smart traffic monitoring applications, automatic license plate recognition, traffic accident detection were examined. Applications focusing on the detection of weather conditions such as cloudy, rainy etc. via video for weather monitoring were examined. Within the scope of sports application area, applications for the detection of actions in different sports activities and smart monitoring of training videos were examined. Within the scope of health application area, smart medical imaging, image-based diagnosis and diagnostic applications, image-based dental health monitoring applications that can be used in the home were examined. Security and public safety applications were examined within the scope of different sub-application areas. Firstly, smart applications developed for the detection and control of natural disasters such as forest fires were examined. Finally, applications developed within the scope of accidents and building safety management within the scope of security and public safety applications were examined.

A YOLO-based smart traffic monitoring approach was developed [22] in a study. It was aimed to increase the performance of the neural architecture to be used by adding Ghost Convolution to the YOLO architecture, and this specialized new neural architecture was named as GC-YOLOv9 by the authors. The mAP@50 evaluation metric of the proposed method was obtained as 77.15%. Video-based violent activity detection is carried out with an artificial intelligence-supported approach was proposed [23]. In the study, first, a CNN process is applied to detect suspicious objects such as knives and guns. Then, video frames containing these suspicious objects are considered in a CNN and LSTM (Long Short-Term Memory)-based neural approach to detect violent activity. It was emphasized that this developed method achieved 3.9% higher performance results compared to existing approaches in the literature.

In another study, an AI-supported live event monitoring approach was developed in smart cities [24]. It was emphasized that as a result of AI-based live event imaging approaches, event-object detection, reporting, analytics and alarm applications were developed in the study. In this context, a neural architecture that performs event and object

detection using CNN and RNN layers together was first developed. Then, a user dashboard was developed showing the results of this neural architecture. In this way, it was aimed to develop a system where the neural architecture outputs obtained from IIoT devices and their data can be manually controlled by city managers. The proposed neural architecture performed the event-object detection process with a success rate of over 98%.

In another approach in the literature, a neural architecture was developed for automatic evaluation of videos in remote patient care procedures within the scope of smart health application [25]. 5G technology was used in the transmission of videos to healthcare personnel. The main contribution aimed to be presented with the study is to eliminate the need for healthcare personnel to manually evaluate all videos thanks to the developed neural architecture. In this context, the developed neural architecture evaluates the accuracy of patient exercises within the scope of remote patient care. The exercise accuracy of the developed neural architecture here is over 90% in binary classification performance.

An approach was developed to increase the reliability of cameras that can perform inference operations with deep learning and to reduce the inference latency [26]. At this point, the authors focused on reducing the computational cost and performing the backup operation in images belonging to different cameras with overlapping fields of view (FoV) with the method they call CrossVision. With this approach, the effective implementation of the coordination and planning of operations in this distributed network of different cameras was ensured. The proposed approach has a multiview feature with all these purposes.

Another study focused on the memory problem caused by the large size of video data in smart cities. In this context, they developed a new method called digital retina for compressing smart city video data [27]. Another study focused on reducing the processing costs in order to increase the inference speed of DNN-based approaches [28]. Another study proposed a new coordination approach to reduce the latency in query-based video imaging approaches such as license plate query [29].

In another study, 3D CNNs were used to perform video shot segmentation and with this approach a new video summarization method was developed for surveillance videos [30]. Firstly, shot segmentation was performed in the developed approach. Then, the obtained video shots were binarized and key-frame detection was performed. The proposed method has a video summarization performance close to 88%.

The focus was on performing crowd density analysis in smart city videos [31] in a study. In the proposed approach, firstly, contrast enhancement of video frames is performed with CLAHE. Then, InceptionV3 CNN architecture is used for feature extraction. Finally, crowd analysis process is performed by using GRU, a recurrent neural network architecture. The proposed method performed crowd analysis process with a success rate over 96%.

The focus was on detecting firearms and human faces by using deep learning methods for the construction of secure smart cities [32] in an approach. FRCNN architectures are used to perform the mentioned detection processes in the proposed method. It was observed that the trained FRCNN performed the detection process with 77.02% success.

In a study, multiview summarization and activity recognition processes were performed for smart surveillance applications [33]. A CNN architecture was used for the shot segmentation process in the developed approach. It was aimed to save computational resources, bandwidth and transmission time thanks to the summarization process.

In another study, traffic accidents were detected from videos recording events on the road and the detected video sections were used in the summary video [34]. The authors aimed to save storage costs thanks to this summary process. The authors used a CNN architecture to perform the accident detection process over video. The developed CNN architecture performs the accident classification process with 82.3% success.

In a literature work, video surveillance applications in the literature within the scope of smart cities were examined [35]. Smart video surveillance applications are examined under the subheadings of smart government, smart home, smart health, smart society and smart environment in the study. The applications developed within this scope are discussed under the subheadings of anomaly detection, criminal incident detection, fire detection and object detection. A YOLO-based smart city monitoring system was proposed in a study [36]. One of the main contributions of the developed approach is that the software architecture it uses allows the neural model used to be trained and updated with new data while the system is operating uninterruptedly. The selected YOLO neural architecture is YOLOv7. The proposed method focuses on detecting 11 objects. The mAP performance of the neural model was calculated as 70.6%. IoT and machine learning approaches in the literature were examined for smart parking applications in a study [37]. In this context, when video-based smart parking applications were examined, it was seen that YOLO and different neural architectures were used. In a study, artificial intelligence-supported smart video surveillance applications were examined for public safety [38]. The main point that distinguishes this study from other literature studies is that it also evaluates video surveillance applications within the scope of policy and regulations. The general scope of video summarization applications was determined by application areas such as object and action recognition, anomaly detection and extraction of meaningful statistical data. An approach was focused on where departments in a smart city infrastructure requested access to and analysis of videos they needed [39] in a study. In another study, an adaptive object tracking approach was developed for videos taken under different lighting conditions [40]. A generalized event processing approach based on deep learning was proposed in a literature work [41]. In the developed approach, traffic management, smart parking and security studies were determined

as the application areas of event processing. Table 1 provides a comparison table of the studies covered within the scope of the literature review. Figure 2 shows the classification tree of video analytics approaches used in smart city surveillance systems. When Figure 2 is examined, it is seen that since the analysis of each frame of the videos is required to perform face recognition and license plate recognition approaches, a summary approach that preserves this information cannot be implemented. In Table 1 there is many approach using deep learning, optimization and some other unsupervised, supervised or semi-supervised technics. Video data belong to smart city applications generally are not labeled. For this reason especially most summarization tasks are performed by semi-supervised approaches using deep learning based features [6], [9], [17]. Addition to this, some approaches uses event or object centric features obtained by deep learning for performing video summarization tasks. While dynamic video summarization techniques are utilized in event-centric video summarization approaches, static video summarization techniques are generally utilized in object-centric video summarization approaches.

However, it is seen that the processing costs of all manual observations made by humans and automatic analyses made by computers can be reduced to a high level with the summarization approaches to be developed for video surveillance applications other than these.

III. MOTIVATIONS AND CONTRIBUTIONS OF THIS STUDY

In this study, as mentioned before, a multi-purpose and effective video summarization method is proposed for smart city video surveillance. When the literature studies given in the previous sections are examined, video summarization approaches are generally handled in a way that preserves the content meanings and are carried out for the purpose of saving time at the user level. As it is known, the videos produced by smart city systems are quite large and again when the literature studies above are examined, they can be used in many different video analytics applications. In addition to the fact that video analytics applications are developed for different purposes, it is obvious that the performance of the approaches with new methods has increased and will increase even more over time. It is quite difficult to handle this large amount of video data as a whole in a new video analytics application in the future and to analyze it manually or automatically for abnormal event-based analyses, and such an approach is not efficient, especially for a concept such as a smart city where sustainability is at the forefront. All these situations reveal the need for summarization of surveillance videos, especially those belonging to smart city systems. As a result, our initial motivations and requirements for focusing on this subject are listed as follows;

- Surveillance videos produced within the scope of smart cities take up a lot of space, and manual or automatic analysis of a lot of unnecessary data is carried out in terms of events and statistical features.

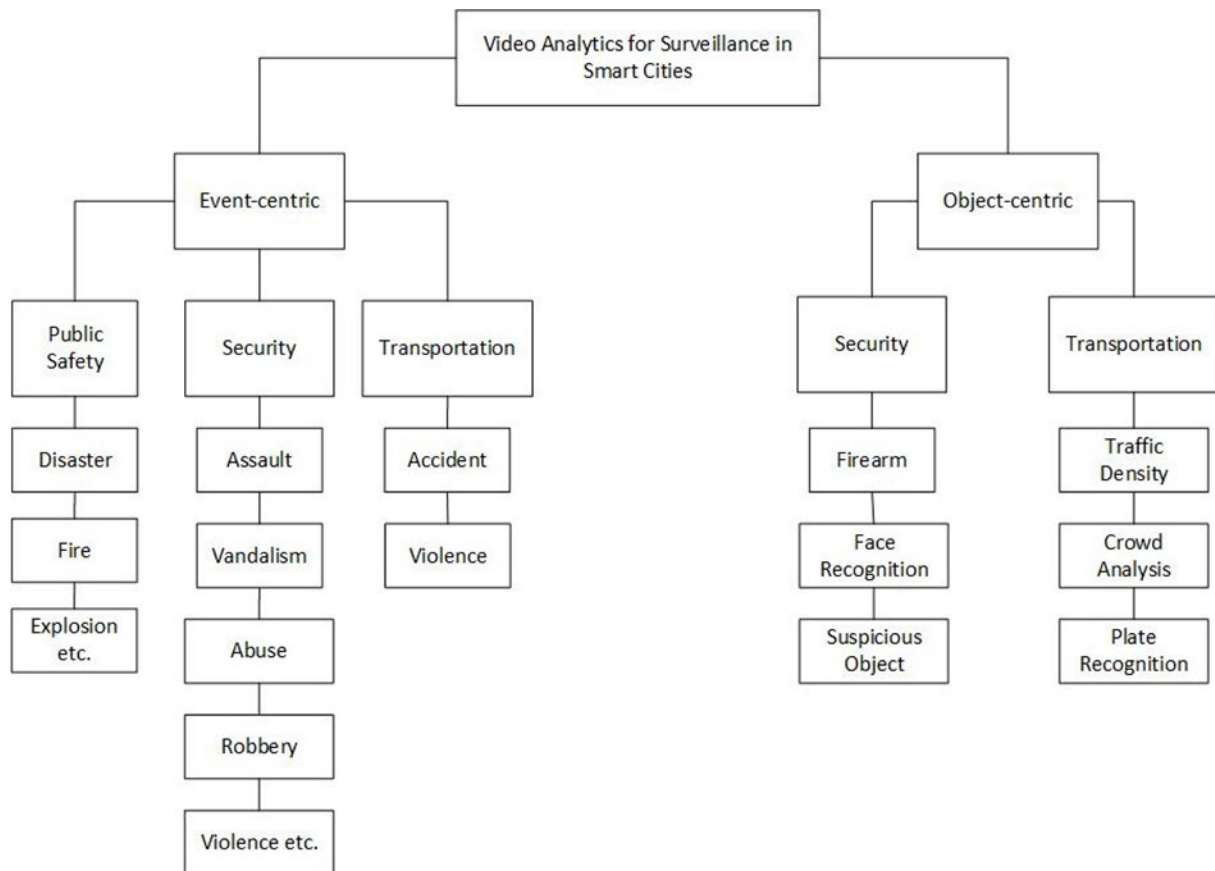


FIGURE 2. Application taxonomy tree of video analytics for surveillance in smart cities.

- Although direct video summarization approaches have been included in smart city video analytics application areas, an approach has not been proposed for summarizing videos produced by video surveillance systems. Generally, the approaches given in this context focus on summarizing videos such as movies, sports activities, online meetings - lessons.
- The purpose for which data summarization approaches are implemented is very important. Since the summarization purposes of even the summarization approaches developed using the same data set are very different, very unrelated summary video outputs can be obtained.
- Existing video analytics approaches need to be examined to determine the purpose for which the smart city surveillance videos will be summarized.
- Data summarization approaches can have very different qualities. For example, modality, single or multimodal approaches, extractive or abstractive qualities can be given as examples. It is necessary to analyze and discuss well which qualities the video summarization approach to be developed should have. In this context, the literature analysis that we shared with you in the previous sections was carried out within the scope of current video summarization approaches and

video analytics in smart cities. As a result of the literature analysis we conducted, it was observed that video analytics approaches in smart cities were generally carried out around the solution of the problems we listed below;

- Applications that perform object detection, tracking and analysis based on these operations on video,
 - Applications that detect abnormal events such as violence, disasters, accidents on video and perform analyses related to these processes,
 - Applications that use techniques such as flow scheduling and edge computing to perform approaches such as machine learning, data mining, and image processing used in video analytics methods at maximum efficiency,
 - Manual observation of abnormal events, especially in the context of people working in urban administration or infrastructure, outside of literature studies.
- The conclusions drawn as a result of examining the applications carried out within the scope of video summarization are listed as follows;
- Neural architectures that can use specialized sequential image frame relationships for video summarization or perspectives where the relationships obtained by

TABLE 1. Comparison of literature studies focused on video summarization and / or video analytic applications in smart cities.

Reference Number	Method	Dataset	Evaluation Metrics	Score	Short Brief
[3]	Image processing Optimization	VSUMM OVP	F1 Score	83.25%	Static Video Summarization
[4]	Neural Encoder - Decoder	TVSum SumMe	F1 Score	60.7%	Keyshot detection based video summarization
[5]	Clustering Regret minimization	OVP	F1 Score	50.59%	Property constrained video summarization
[6]	Deep learning	DPCS	F1 Score	78.78%	Multimodal approach for sports video summarization
[7]	Deep learning	TVSum SumMe	F1 Score	77.7%	Adaptive video summarization approach using object behavior detection
[8]	Deep learning	-	F1 Score	98.7%	Video summarization by removing no-human frames.
[9]	Deep learning	TVSum SumMe	F1 Score	45.89%	Topic-aware multimodal video summarization
[10]	Deep learning	Self annotated badminton tournaments video	F1 Score	46.99%	Event-centric video summarization
[11]	optimization k-means quick sort	YouTube	-	-	Video summarization approach using different ML technics
[14]	Deep learning	MSR-VTT	ROUGE-1 ROUGE-2 ROUGE-L	40.21 17.64 36.89	Abstractive video to text summarization approach
[15]	Deep learning	-	ROUGH	0.56	Abstractive video to text summarization approach
[16]	Deep learning	Switchboard	ROUGE-1 ROUGE-2 ROUGE-L	0.40 0.16 0.32	Abstractive video to text summarization approach
[17]	Deep learning	VSUMM TvSum SumMe RAI	F1-Score	0.91	Deep learning based keyframe selection approach.
[20]	Deep learning	[42]	F1-Score	67%	Performance analysis of different object detection models for smart city video analytics.
[22]	Deep learning	BDD100K Cityscapes	mAP@0.5	77.15%	A YOLO based smart city monitoring approach
[23]	Deep learning IoT	RWF200	F1-Score	80.0%	Deep learning based violence detection approach performed in IoT edge.
[24]	Deep learning	YouTube	F1-Score	97.77%	A deep learning based event monitoring approach in smart cities.
[26]	Deep learning	EPFL AI City SALSA	F1-Score	96.5%	A deep learning based object detection approach performed on IoT edge.
[28]	Deep learning	BDD100K	-	-	Effective model inference approach for decreasing latency in video IoT systems in Smart Cities.
[29]	Deep learning	RetineNet-50	Analytic Accuracy	35.7%	Effective model inference approach for decreasing latency in video IoT systems in Smart Cities.
[30]	Deep learning	YouTube TVSum	F1-Score	69.9%	Deep learning based video data prioritization approach
[31]	Deep learning	-	F1-Score	93.60%	Deep learning based crowd analysis approach for smart cities.
[32]	Deep learning	Internet Movie Firearms Database	mAP@0.5 mAP@0.75 mAP@0.5-0.95	77.02% 16.40% 29.73%	Deep learning based firearms detection approach for secure smart cities.
[34]	Deep learning	Syntetich dataset	F1-Score	87.1%	Deep learning based Event-driven traffic monitoring approach.
[36]	Deep learning	COCO2017 KITTI	mAP@0.5	87.8%	YOLO based real time traffic monitoring approach.
[40]	Deep Learning IoT	OTB2013	MPAT	82.4%	A deep learning based efficient object tracking approach.

classical image processing and analysis techniques are handled with different data mining or machine learning approaches can be used.

- With multimodal neural architectures, in addition to video data, audio data can also be used within neural architectures.

- Features such as object detection - tracking, activity recognition, histogram analysis can be used in video summarization. When neural architectures are not used for summarization, data mining and machine learning approaches such as clustering, optimization, etc. are used.

- The implemented video summarization approach can be multiview in nature, especially for an application area such as sports videos.

- Summarization approaches can be based on keyframe detection or keyshot selection.

The features and contributions of the video summarization application we developed in light of the literature research results presented above are listed as follows;

- The video summarization approaches we developed within the scope of smart cities need to provide summary videos with two different qualities, consisting of abnormal event sequences and preserving object detection and tracking features.
- Multiple neural architectures need to be developed for the event-based video summarization approach and compared according to their memory, computational cost and accuracy criteria. In this context, LSTM or attention layers need to evaluate these features in order to extract image features and learn sequential relationships with pre-trained and unpre-trained CNN architectures.
- The trained models need to classify the video sequences as binary normality - anomaly, and the anomalous video sequences are included in the summary video. Therefore, the summary performance should be evaluated with the classification performance of the relevant models.
- There is no need to classify abnormal events within themselves in the classification process, because in the summarization process, the main criterion is that the event is abnormal, not its type.
- In video analytics approaches that use object detection and tracking, the main purpose is usually to perform these operations or to solve problems such as traffic density, crowd analysis, etc. as a result of these operations.
- The approach that best preserves information about this process is heat maps. With heat maps, information about detected objects and their movements is visualized with maximum preservation.
- Most smart city video surveillance systems currently perform object detection or tracking on the edge or cloud. For such approaches, an approach should be used in which the heat maps obtained as a result of object detection are used in the video summarization process and n frames similar to this heat map are selected with optimization when combined.
- For an approach that does not currently perform object detection or tracking, the same optimization process should be performed by creating heat maps based on optical flow analysis.
- In order to evaluate both approaches, the number of objects detected in the original video and the summary video should be compared. In addition, the parameter setting used in the optimization process should be performed according to the results obtained from these comparisons.
- As can be seen, the object detection-focused summarization process is a summarization approach based on

keyframe selection as a feature. The event anomaly-based summarization process is a summarization approach based on keyshot determination as a feature.

- Both approaches do not require a multiview or multimodal feature. Since the output of the developed approach is video, the summarization process is extractive.
- For example, reporting a video from video analytics approaches requires abstraction or multimodality. Our aim is to produce summary videos for low-cost integration of these approaches into smart city systems. Therefore, multimodality was not needed.
- The developed approaches do not currently provide multiview summarization, but a smart city surveillance system already consists of more than one imaging device. In order not to lose important features or events, it would be a more accurate approach to summarize the video from each imaging device separately.
- Apache Kafka technology should be used to integrate the developed approaches and to ensure adaptive integration in cases where they will work on the edge and cloud.

The contributions of the proposed method are listed as follows;

- This summarization method developed for video surveillance in smart cities does not only preserve valuable information for manual surveillance, but also ensures that valuable features are preserved in summary videos in terms of video analytics applications performed in literature and existing smart city systems.
- The performance of the attention layered Transformer model developed in the event anomaly based video summarization approach is over 95% (96.7%), and the memory and processing cost required for the inference process of this model is also quite low.
- In order to select the image frame representing the most important moment of the event from the video parts that are considered abnormal, attention scores calculated in the Transformer model were used. This developed approach is quite innovative and is quite efficient in terms of not performing a new model training specifically for the keyframe detection process.
- In the object detection-tracking based video summarization approach, the selection of keyframes is performed by an optimization process based on the similarity of heat maps.
- While the object counting process performed for the evaluation of the summarization approach, in which heat maps are obtained as a result of the object detection process, has an average success of 93.4%, the average success obtained as a result of the evaluation of the optical flow process is 92.7%. This shows that the obtained summary videos preserve their statistical properties with a very high performance.
- As a result, in this study, the main requirements for multi-purpose summarization of smart city videos were determined, different approaches that meet these

requirements were presented and compared, and Apache Kafka technology was used to facilitate integration into all smart city systems. Kafka was primarily chosen because it provides a system that can store and replay data streams. Flink and Spark are generally transaction-oriented and have limited data storage capabilities. With the developed method, time losses in manual observation of event-based videos, processing costs in automatic analysis, and processing costs of video analytics applications focused on object detection and tracking were minimized. Moreover, thanks to the developed method, the original video can be deleted from the hard disk with decisions to be taken within the scope of policies to be determined by the smart city management, thus preventing unnecessary storage costs.

- These advantages of the proposed method and minimizing processing, storage and memory costs provide additional important contributions to smart cities, especially within the framework of sustainability.

IV. PROPOSED APPROACH

In this study, as mentioned before, a new and multi-purpose video summarization approach has been developed for video surveillance applications in smart cities. In this approach, it is aimed to obtain two different summary videos for event-centric and object-centric video surveillance applications. In this context, two different summary approaches have been developed. The first of these is a video summarization approach using genetic algorithms to be used in feature or object detection/tracking-centered video analytics approaches and is realized by merging n keyframes selected for maximum preservation of features. In the second approach, a video summarization approach is created by merging abnormal video sequences by using a neural architecture where video sequences are classified as normal / abnormal. The block diagram expressing the entire video summarization approach developed is given in Figure 3. When Figure 3 is examined, it is seen that real-time videos received with IoT edge are taken to the video analytics queue with a Kafka stream. In smart cities, calculations can be performed on the edge or cloud in video analytics approaches. Apache Kafka is a current and very important component in video analytics applications for effective processing of video streams. Therefore, it is aimed for the developed approach to work in an integrated manner with Apache Kafka. The two basic summarization components of our approach will consume the video batches in Kafka topics separately. In fact, the approach that performs feature-centered summarization process focuses on two different scenarios. As it is known, object detection and/or tracking operations are frequently performed on the edge in video analytics applications performed for surveillance applications in smart cities. Therefore, if the video summarization approach we developed is integrated into such a system, using a heat map created depending on the object detection performed on the edge and addressing it in the genetic algorithm will only load the cost of the

genetic algorithm operation to the current system. However, if our video summarization system is integrated into a smart city video surveillance system that does not currently have object detection capability, this process will be performed with heat maps obtained by optical flow analysis in order to minimize the processing cost. The purpose of the genetic algorithm is to ensure the selection of a certain number of frames that will obtain the heat map closest to the heat map representing the entire video. In the second approach, training of more than one neural architecture that performs the video classification process was performed. In the video classification process, an approach that can learn the relationships between frames in addition to visual features should be used. Therefore, pre-trained and unpre-trained CNN architectures were used to extract visual features in the first layers of neural architectures. Attention or RNN-based neural architectures were used to learn the relationships between frames by using these features. Multiple neural architectures were trained with different configurations of these options and compared according to their features such as performance, loss, and number of parameters. Again, as seen in Figure 3, in cases where the classified video sequence was abnormal, the video summary process was performed by ensuring that these sequences were included in the summary video. It was deemed useful to examine these approaches under separate headings in order to understand their features such as detailed explanations and evaluation metrics of both developed approaches.

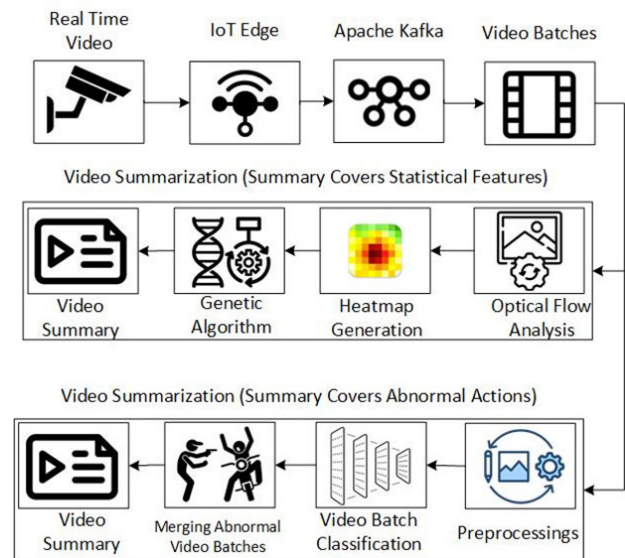


FIGURE 3. An overall block diagram of the proposed approach.

A. OBJECT-CENTRIC VIDEO SUMMARIZATION

For object-centric video summarization, two different scenario-adaptable approaches have been developed according to the current features of the smart city system to which the system will be integrated. The first of these focuses on

smart city systems that currently perform object detection or tracking operations on the edge. In this approach, a heat map of the original video should be obtained by using the results of the object detection process. After this process, the genetic algorithm selects n frames in the original video using the structural similarity (SSIM) of the heat map created by combining these frames to the heat map of the original video. In this way, the system only bears the processing cost brought by the genetic algorithm for the video summarization process. In the second case, smart city scenarios where object detection is not currently performed are considered. In this case, the heat map of the original video is created based on the results of optical flow analysis. After obtaining the reference heat map, the heat maps created by the image frames selected by the genetic algorithm are also performed by means of optical flow analysis. Optical flow analysis has a lower processing cost compared to object detection. Figure 4 shows a block diagram showing the general flow of the event-centric video summarization module. As we mentioned before and also as seen in Figure 4, heat maps were used in both approaches. The following steps were followed in the heat map created based on object detection;

- The empty matrix of the heat map is created in frame size.
- Information on detected objects was taken into consideration one by one.
- In order to update the matrix, the value of the pixels in the area covered by each object is increased by 1.
- This heat map matrix is divided by the number of frames in the video to normalize the heat map.

In other words, the heat map method obtained provides a method that analyzes the motion and location density from the object detection outputs. Using the detected object coordinates in each frame, the values of a density matrix in frame dimensions are updated. The heat map is normalized according to the number of frames and creates a density model that represents the overall motion and object distribution of the video.

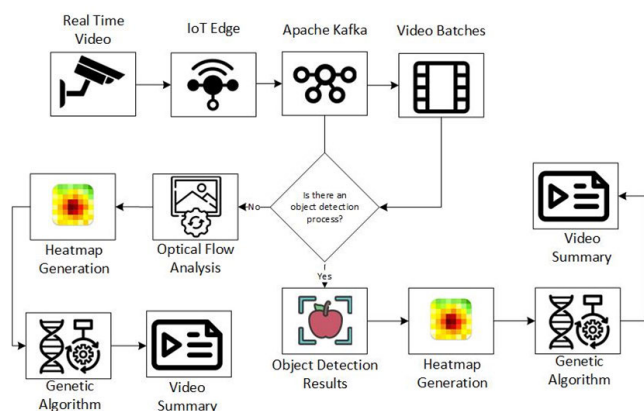


FIGURE 4. Block diagram of the object-centric video summarization approach.

In cases where there is no object detection, optical flow analysis was used as mentioned before. Optical flow calculations allow analyzing spatial and temporal motion intensities by determining pixel-level motion between consecutive frames. This process is a critical preprocessing step in video analytics applications such as motion-based anomaly detection and behavior analysis. For the optical flow process, the image was first converted from RGB to gray. After this process, the steps of optical flow analysis are explained with the following mathematical expressions;

$$F(x, y) = \begin{bmatrix} v(x, y) \\ u(x, y) \end{bmatrix} \quad (1)$$

$$M(x, y) = \sqrt{v(x, y)^2 + u(x, y)^2} \quad (2)$$

$$\frac{\partial I}{\partial t} + u \frac{\partial I}{\partial x} + v \frac{\partial I}{\partial y} = 0 \quad (3)$$

Figures that are meant to appear in color, or shades of black/gray. Such figures may include photographs, illustrations, multicolor graphs, and flowcharts.

Figures that are composed of only black lines and shapes. These figures should have no shades or half-tones of gray, only black and white.

When the given equivalence is examined (1)-(3);

- $F(x,y)$: Optical flow vector in (x,y) coordinates,
- $U(x,y)$: The optical flow component representing horizontal motion,
- $V(x,y)$: The optical flow component representing the vertical motion,
- $M(x,y)$: The magnitude of movement in the (x,y) coordinates,
- $I(x,y)$: Pixel value in (x,y) coordinate of image I .

The optical flow analysis approach used here creates an optical flow heat map for the entire video. In the object-centric summarization approach developed for both scenarios, separate heat maps are created for the individuals in the genetic algorithm population. Then, SSIM calculations are performed for the heat map of the entire video and the heat maps of the individuals. In this way, the fitness value of each individual is calculated. Figure 5 shows the steps of the genetic algorithm approach that performs the summarization process. It is useful to explain the steps in Figure 5 separately. First of all, the individuals in the population have genes equal to the total number of frames in the video. Each gene holds the information about whether the frame it represents will be selected or not, so there is a binary encoding situation. In genetic algorithm terminology, individuals are called chromosomes and chromosomes coded in this way are binary encoded chromosomes. When creating the initial population, care is taken to ensure that the number of frames to be selected is equal to the maximum number of frames. The randomly generated chromosomes for the beginning are equal to the maximum population size. The crossover step is used in genetic algorithms to ensure genetic diversity. In our approach, random chromosomes are selected for crossover and gene changes are performed by generating

random numbers with a crossover ratio of 0.5. If the number of selected frames is more than the maximum number of frames, the selection value of some randomly selected frames is equalized to 0. The mutation process is performed with a 2% probability and if this condition is met, it is provided by taking the logical expression held by the relevant gene (if 1, 0, if 0, 1). In cases where the maximum number of selected frames is exceeded, the number of selected frames is reduced by the method used in the crossover process. After these operations, the heat map of each chromosome and the heat maps of the original video are evaluated with the SSIM metric. SSIM is used to calculate the structural and perceptual similarity between two images. It results in a similarity score by taking into account the brightness, contrast and structural information of the images. This metric is widely used especially in video and image processing fields, compression, image quality assessment and restoration processes. The mathematical expression of the SSIM metric is given below [43].

$$SSIM(x, y) = l(x, y)^\alpha \cdot c(x, y)^\beta \cdot s(x, y)^\gamma \quad (4)$$

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_1}{(\sigma_x)^2 + (\sigma_y)^2 + C_1} \quad (5)$$

$$l(x, y) = \frac{2\mu_x\mu_y + C_2}{(\mu_x)^2 + (\mu_y)^2 + C_2} \quad (6)$$

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \quad (7)$$

When the given equivalence is examined (4)-(7);

- $l(x, y)$: Compares the average brightness of two images,
- $c(x, y)$: Compares the contrast levels of two images,
- $s(x, y)$: Compares the structural order of two images,
- α, β, γ : Determines the importance weight of each component, usually all three values are taken as 1
- μ_x : The average brightness of the image x,
- μ_y : The average brightness of the image y,
- C_1 : A small constant used to prevent division by zero,
- σ_x : Standard deviation of image x (contrast), σ_y : Standard deviation of image y
- C_2 : A small constant used to prevent division by zero,
- σ_{xy} : Covariance between x and y (structural order similarity)
- C_3 : Constant.

After calculating the fitness values of each chromosome, the tournament method was used for selection. The processes mentioned above are repeated as many times as the generation number. Basically, the generation number determines how many iterations the genetic algorithm will work for.

In order to evaluate the object-centric video summarization approach we developed using the genetic algorithm, we compared the number of objects in the original video and the summary video. The number of objects detected in the summary video was divided by the summary video frame number and the value obtained from here was multiplied by the original video frame number. This final value was calculated by taking the number of selected moving objects such

as vehicles in the original video as reference and performing the accuracy calculation.

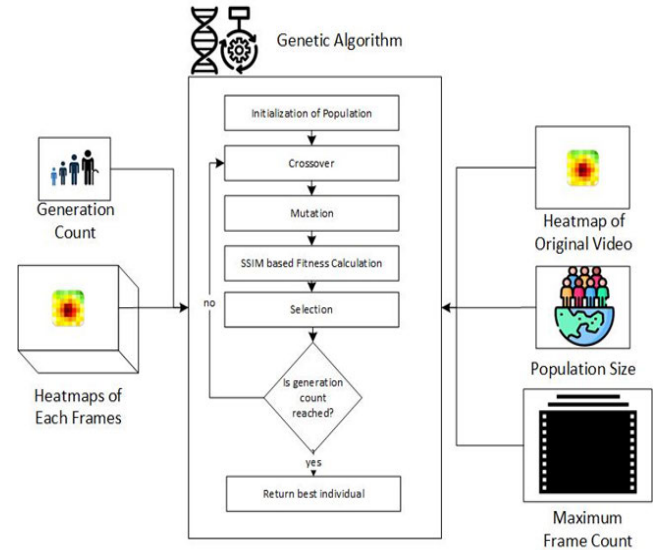


FIGURE 5. Block diagram of the genetic algorithm process in the object-centric video summarization approach.

The reason for choosing such an evaluation method is to reduce the processing cost of video analytics applications that perform statistical analyses based on object detection, which is the reason why we chose such an approach for summarization. The use of heat maps in the optimization performed actually ensures that the method we present is not only preserved in terms of the number of objects but also the features related to local object densities and object movements. Considering that object detection and tracking are at the core of many data analytics applications, the selected evaluation metric is also seen to be quite valuable metrics for determining the summary quality of the fitness (SSIM) value calculated in the genetic algorithm stage of the summary video. Finally, we are aware that the system we developed is not an adaptable approach for some of the object-centric video analytics applications we have given in Figure 2. The first reason for this is that operations such as face or plate recognition need to check all frames of the video. Therefore, face or plate recognition operations performed on the summary video will lead to an inadequate and illogical approach. The second reason is that the video analytics application to be performed in operations such as suspicious object detection has a dependency on the object class, and since these are abnormal objects, it is not possible to present this with an approach that aims to generalize the video like ours. This deficiency of our object-centric summarization approach will be completed with our event-centric summarization approach. Apart from this, as mentioned above, the aim of the developed approach is to present a video summary that generalizes the video. For an intelligent video analytics focused on statistically detecting abnormal situations, a reference heat map was obtained by averaging the heat maps of all frames in

the videos based on optical flow or object detection. The n frames with the farthest structural similarity to this heat map were given as the abnormal keyframes of the video as an output of our approach. The most normal and abnormal keyframes obtained with this approach will also be included in the experimental results.

B. EVENT-CENTRIC VIDEO SUMMARIZATION

The goal of our previous summarization approach was to preserve valuable features in the video for video analytics applications based on object detection or tracking. Our event-centric video summarization approach appeals to both video analytics applications focused on event-based analysis and smart city workers performing manual event monitoring. In addition, we wanted to implement both of our summarization approaches in a way that would cover manual or computer-aided surveillance applications in smart cities. The applicability of the object-centric video summarization approach we presented above is limited for some object-centric approaches within the scope of video analytics. For example, in some object-centric approaches, operations such as fire detection, gunshot detection, and suspicious object detection are performed on the image. In other words, the purpose of the analytical applications performed here is not to obtain statistical results but to detect abnormal situations. When you examine the characteristics of the videos we used under the scope of training and test data in the experimental results, you will see that the Event-centric video summarization approach we will develop will also eliminate the deficiencies of the object-centric video summarization approach for video analytics applications focused on abnormal object detection. If we proceed with a specific scenario, the detection of objects such as knives, which can be used as a tool of violence for anomaly detection and in the normal flow of life, is misleading with an object-centric approach. However, since the videos are examined as streams rather than single frames with the approach we present here, analyses can be performed on sequences where these suspicious objects are actually in suspicious situations. Therefore, with the event-centric video summarization approach, which is the second component of our main system, we present an integrated system that addresses all video analytics applications performed within the scope of video surveillance in smart cities, except for continuous video analytics applications such as face or license plate recognition. Since the output we will present in our proposed approach is a summary video containing anomalous situations, we did not want to complicate the solution. Therefore, we evaluated the video segments with this architecture by using a neural architecture that performs binary classification and realized the presence in the summary video in this way. A block diagram of the proposed event-centric video summarization approach is given in Figure 6. When Figure 6 is examined, the basic process is the classification of video batches in the Kafka queue with a neural architecture for a certain period of time (in our case, 3 seconds). At this point, since it would be unnecessary to

evaluate all frames belonging to videos taken in a certain period of time in the neural architecture, frames should be selected from the video batch and handled by selecting them at homogeneous intervals.

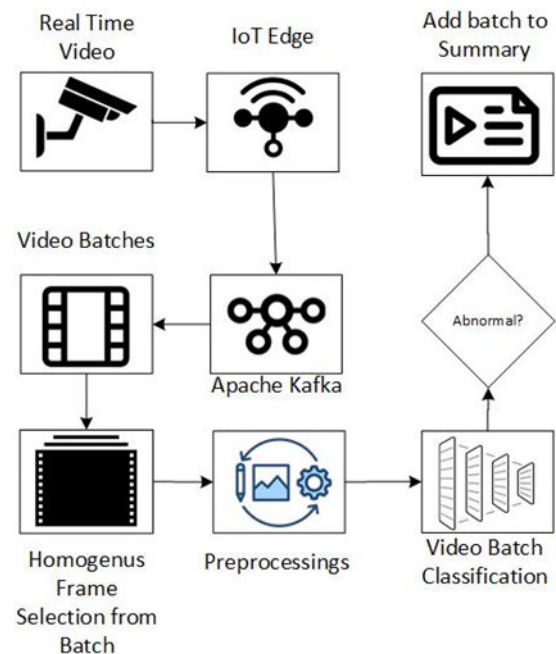


FIGURE 6. Event-centric video summarization flow.

In the preprocessing step, it should be ensured that the frames are of a size suitable for the input of the neural architecture. More than one neural architecture was developed for the video batch classification process, where the entire node is solved. These were then compared based on different criteria, which you will see in the experimental results section. The stages you see above actually only show the inference stage of a trained neural architecture. In order to fully understand the qualities of the developed neural architectures, the training process needs to be explained in detail. Figure 7 shows the training processes of the neural architectures developed for the classification process. More than one dataset with video content considered abnormal was used for the training process. The quantitative and anomalous content information of these datasets is given in detail in the experimental results. Since the classification process to be performed will be used in a summarization application, using the raw version of the videos in the dataset is not a correct approach. Because most of the videos in the dataset consist of normal video sequences in addition to the abnormal sections. Since the classification to be performed by the neural models we trained is used in the summarization process, the normal sequences in the abnormal videos need to be cleaned from the video. In fact, evaluating these cleaned normal sequences with the normal label in the training data will significantly increase the ability of the neural architecture to distinguish normal and abnormal situations in the same video. A computer-aided manual

approach was used to fulfill these requirements explained above in the datasets. In this approach, each video in the dataset was shown as three-second sequences, and normal video sequences were labeled as normal, and abnormal video sequences were labeled as abnormal. While selecting the sequences, care was taken to ensure that the normal and abnormal sequences selected from each video did not have a very heterogeneous distribution. Again, a few sequences of maximum three seconds from the normal videos were labeled and included in the training data. Quantitative information about the final dataset used for training is given in detail in the experimental results section.

The neural architecture that will implement such an approach must have different capabilities. Since the incoming video data is basically a sequence of images, it must first obtain the visual features of these frames with CNN architectures. If the classification process were performed on a single image, the features extracted by CNN architectures could be subjected to the Flatten process and the remaining processes could be completed with an FCL (fully connected layer) structure, but in our case, there are multiple images that have temporal dependence with each other. Therefore, the features obtained from CNN architectures must be handled with a neural architecture that can learn the relationships between multiple elements in the same structure. Despite this, we have included the training results of neural architectures created only from CNN and FCL architectures in the experimental results. When it comes to learning the relationships between data with temporal dependence, the first things that come to mind are RNN architectures and attention layers. Therefore, we also trained neural architectures where RNN and attention architectures are used separately. Finally, as it is known, transfer learning approaches are used in the training of neural architectures, especially when the data set is limited. We also performed neural training with this approach by adding pre-trained neural architectures to the CNN layers of some of our models. As can be seen, configurations consisting of different architectures can be used to perform the classification process we need within the scope of deep learning. Therefore, before talking about the structure of the classifier neural architectures that we will compare, it was deemed appropriate to explain concepts such as CNN, transfer learning, LSTM, and Attention in detail.

The purpose of CNN architectures is to improve the ability to learn features in visual data in a hierarchical manner. A basic CNN architecture includes convolutional layers, pooling layers, and fully connected layers. In convolutional layers, feature extraction is basically performed on the image being considered. Filters are used in these layers to learn features such as edges, textures, etc. in an image. Filters are shifted over the image to create feature maps for the image.

Pooling layers reduce the computational cost of the model by preserving high-value features from feature maps, and average and maximum pooling layers are usually used. In fully connected layers, there is an architecture consisting

of classical neurons. This layer, which handles the features coming from convolutional and pooling layers, is located at the end to perform the final task of the model. For example, a fully connected layer with five neurons should be in the last layer of a model that performs classification for five different classes. Intermediate fully connected layers can also be found between the last pooling layer and the output fully connected layer of the CNN architecture to increase the learning ability. Another very important concept related to CNN architectures is the transfer learning process. Especially in the training processes of quantitatively insufficient data sets, the feature extraction capability of convolutional layers trained from scratch on images may not reach a sufficient level. In such cases, CNN architectures such as AlexNet, VGGNet, ResNet and Inception, which were previously trained with very large data sets, are separated from their fully connected layers and used to create convolutional layers of another CNN architecture. In the training process here, the value of the learned filters can be kept constant and training can be performed only for fully connected layers of the neural architecture or for different layer groups if any. On the other hand, the existing values for the pre-trained learning parameters of the convolutional layers can be used only at the beginning of training and the learning parameters of these layers can be updated in the fine-tuning process. In problems where video data will be handled with CNN architectures, these layers need to be configured in a Time Distributed form. In this way, features of more than one frame of the video can be extracted. Especially in scenarios where video data is handled in a neural architecture, it may be difficult to obtain high-performance results if the features extracted by the convolutional layers are processed only in fully connected layers, depending on the problem. Therefore, when other studies in the literature are examined, these features learned from convolutional networks are also handled with RNN or attention mechanism architectures with high ability to learn sequential structures, and thus, the problems where video data is the input data are solved. This is why we felt the need to train more than one neural architecture and compare their results. Therefore, we also see the benefit of talking about RNN architectures and attention mechanisms. RNN architectures should have a cyclic approach that is fundamentally different from classical neural architectures. In this way, RNNs store information from previous steps in sequential data and use this information in subsequent steps. There are two most commonly used types of RNN approaches. These are LSTM and GRU (Gated Recurrent Units) architectures. As we mentioned before, we used LSTM architecture together with CNN architectures in one of the neural architectures we developed. LSTM architectures basically consist of three structures. Forget gate is used to determine which information will be forgotten. Input gate is responsible for adding new information to the cell. Output gate controls the transfer of the cell state to the next state. In this context, the mathematical representation of the operations performed by

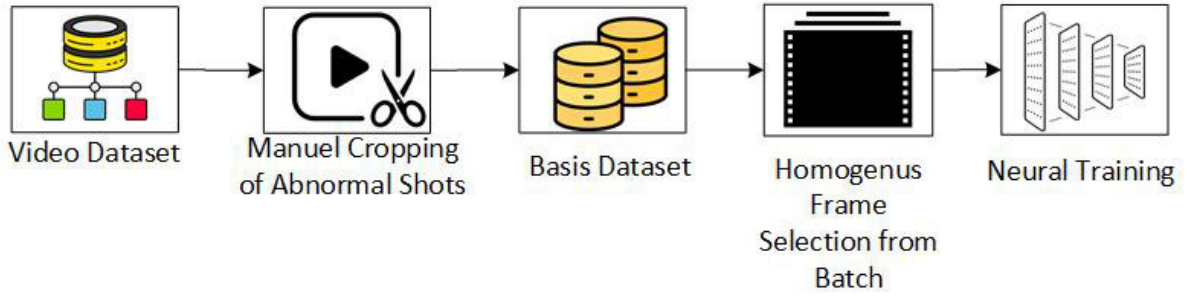


FIGURE 7. Steps of neural model construction.

forget gate, input gate and output gate is as follows.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (9)$$

$$C_t \sim \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (10)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot C_t \sim \quad (11)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (13)$$

When the given equation is examined, it is shown how the C_t (cell state) and h_t (hidden state) information in each LSTM cell is calculated. In all equations, σ is the sigmoid function, $\tanh$ is the hyperbolic tangent function. f_t is the forgetting decision, W_f is the weight, and b_f is the bias. With the calculated forgetting decision, which part of the cell state will be preserved is calculated. i_t is the entrance gate decision, $C_t \sim$ is the cell state candidate, and C_t is the updated cell state. o_t is the output decision. x_t is the input vector. The advantages of LSTM over other RNNs can be expressed by learning long-term dependencies, preserving gradients, and the flexible memory mechanism. In the literature, apart from video processing, it is used for natural language processing, time series estimation, and the analysis of medical signals such as ECG or EEG. Another neural element that needs to be explained is the Attention mechanisms. Thanks to the attention mechanisms, different weights are assigned to different parts of the input data, allowing the model to focus on more critical information. In this process, the model processes a series of inputs $(x_1, x_2, \dots, x_t)$. An importance score is calculated for each input element. The scores are normalized with the softmax function and converted to weights. The weights are multiplied by the relevant input elements to obtain the output of the model. In the attention terminology, three elements called Key (K), Query (Q) and Value (V) are very important. Query expresses the information the model wants to learn. Key expresses the information represented by the input elements. Value expresses the actual values of the input elements. In addition, in the attention score calculation process that we will give below, a size factor used for normalization, expressed as d_k , is used.

T expresses the transposition process. There are different attention types such as Bahdanau, Luong, Self-attention and

Multihead attention according to the usage approach. The most well-known neural applications are transformers. Transformers use a pure attention mechanism, eliminating the need for sequential architectures such as RNN and LSTM.

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (14)$$

In light of all this information, we believe that it is quite appropriate to develop a video classifier for event-centric summarization using CNN and transformer architectures together. In an approach like ours, using attention mechanisms that perform operations by revealing important parts of a data for abnormal and normal video classification will also provide the importance score of each frame. Despite this, it is important for the quality of our study to train different neural architectures and make comparisons. Therefore, it was decided to train four neural classifiers whose characteristics are expressed as follows;

- A neural architecture consisting of nonpre-trained time-distributed convolutional layers and LSTM layers (CNN-LSTM)
- A neural architecture consisting of pre-trained time-distributed convolutional layers and LSTM layers (PRECNN-LSTM)
- A neural architecture consisting of nonpre-trained time-distributed convolutional layers and transformer blocks with attention (CNN-TRANSFORMER)
- A neural architecture consisting of pre-trained time-distributed convolutional layers and transformer blocks with attention (PRECNN-TRANSFORMER)

In neural architectures where pre-trained convolutional layers are used, VGG19 neural architecture was used. Figure 8 shows the diagrams of the trained models. The given diagrams represent the neural architectures in the inference phase.

C. SYSTEM INTEGRATION WITH APACHE KAFKA

Attention scores were not expressed as an output during the training of neural architectures with transformer blocks, but attention scores were also used in the output configuration of the model when these architectures were compiled before inference.

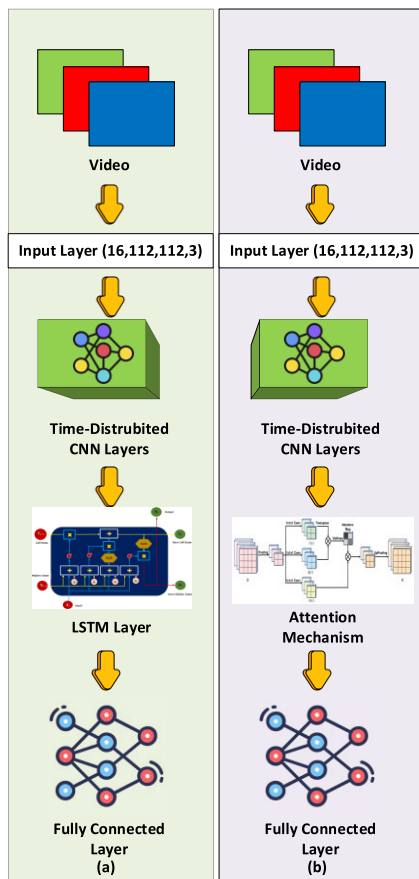


FIGURE 8. Block diagram of the neural models. (a):CNN-LSTM and PRECNN-LSTM, b:CNN-Transformer and PRECNN-Transformer).

Apache Kafka is a stream management technology used for a distributed message queue and real-time data streaming, and is frequently used to increase system efficiency, especially in IoT applications that involve streaming data. In this study, we use Kafka to efficiently manage video streaming and to provide integrated integration of our two different summarization approaches. Before explaining our integrated system, it is useful to explain some concepts used in Kafka terminology.

- **Producer:** Applications that send data to Kafka. Examples include data from sensors, log files, or in our case, raw image frames.
- **Consumer:** Applications that receive data from Kafka and process it. In our case, our summarization applications can be expressed as consumers.
- **Broker:** These are servers running in a Kafka cluster. They enable the transportation, storage and scaling of data.
- **Topic:** These are the data categories in Kafka. Producers send data to a specific Topic. Consumers receive data from a topic.
- **Partition:** Kafka allows parallel processing of data by dividing topics into multiple partitions.

- **Offset:** It refers to the sequence number of each message in the partition.

Figure 9 shows a block diagram representing our video summarization system integrated with Apache Kafka. To test the integrated system, Kafka was installed on a Raspberry Pi 4 and videos taken from the used datasets were combined and published via Raw Video Topic. Consumer components ran on a laptop and consumed the frames coming from Raw Video Topic. While consuming the frames taken from the video topic, video batches to be summarized were determined according to the given frame or time constraints before the summarization process started. After the summarization process of each video batch was completed, the resulting summary videos were named and recorded with the help of the offset numbers belonging to the beginning and ending frames of the video batches. All training operations have been ran on Google Colab platform which enables A100 GPU, its RAM has 80GB capacity and GPU memory has 40 GB capacity. The laptop which acts as the cloud has 2.8 GHz CPU and 6GB NVidia GPU.

V. SYSTEM EVALUATION AND DATASET

A. SYSTEM EVALUATION

Since the approach we developed is multipurpose and modular, each component needs to be evaluated separately and the performance of the integrated system needs to be evaluated separately. Classical traffic or indoor CCTV imaging videos should be used for the evaluation of the object-centric video summarization module. In this context, our primary goal is to obtain the same or over 90% close results when the results of the video analytics approach focused on obtaining statistical features on the original video are applied to the summary video. In this context, an evaluation approach based on object counting, object-related heat map acquisition and comparison of local object densities within the frame will be used on the videos in the data sets given below. In the object-centric video summarization process performed with object detection, since the fitness score calculated by the genetic algorithm is calculated with the structural similarity of the object-detection-related heat maps (H) of the original and summary videos, the first evaluation of this approach will be realized while the summary video is being produced. In this way, the general statistical information protection performance of the video will be evaluated (GSCP). In the evaluation technique where the number of objects is the criterion, the object numbers detected on the summary and original video frames are normalized (N) according to the frame numbers and evaluated. The summary video object number accuracy obtained as a result of the evaluation is expressed as the statistical accuracy of the summary (SAS). This process has been applied as a common evaluation technique for approaches using both optical flow and object detection based heat maps. Finally, the frame numbers in the summary and original video will be compared and the rate at which the video is summarized will be calculated, and this rate will show the video summarization efficiency (VSE). The

mathematical representation of the evaluation metrics used in the object-centric summarization process is given below.

$$GSCP = SSIM(H_{original}, H_{summary}) \quad (15)$$

$$SAS = \frac{|N_{original} - N_{summary}|}{N_{original}} \quad (16)$$

$$VSE = \frac{F_{summary}}{F_{original}} \quad (17)$$

The performance of the event-centric video summarization process is basically directly related to the performance of the classification model. Classical evaluation metrics in machine learning will be used to evaluate the model. In order to evaluate the event-centric video summarization approach performed with this model, videos will be fragmented according to the durations covered by the model. The entire video will be expressed with binary coded vectors (V) marked as normal and abnormal with the dimensions of this number of fragments. Then, the entire video will be subjected to the classification process and the same vector will be produced by the summarization component. As a result of the evaluation of both vectors with MSE and F1-Score metrics, summary loss (SLOSS) and accuracy (SACC) information will be obtained. For SACC, recall and precision values are calculated using true positive (number of elements marked as abnormal in both vectors at the same time, TP), false positive (number of elements marked as abnormal in the summary vector but normal in the original vector, FP) and false negative (number of elements marked as normal in the summary vector but abnormal in the original vector, FN) values. Then, the F1-score metric is determined with the help of these values. Finally, the optimum number of video frames consisting of abnormal sections and the frame numbers of the obtained summary video will be compared to calculate the event-centric summarization efficiency (EVSE). Addition to this, VSE metric was also used for the evaluation. The mathematical representations of the metrics used for the evaluation of the event-centric summarization module are as follows.

$$SLOSS = \frac{1}{n} \sum_{i=1}^n (V_{original,i} - V_{summary,i})^2 \quad (18)$$

$$SACC = F1(V_{original}, V_{summary}) \quad (19)$$

$$Precision = \frac{TP}{TP + FP} \quad (20)$$

$$Recall = \frac{TP}{TP + FN} \quad (21)$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (22)$$

$$EVSE = \frac{F_{summary}}{F_{original_abnormal}} \quad (23)$$

For the integrated system evaluation, the number of frames sent to the Kafka topic by the producer and the number of frames consumed by the consumers will be compared at certain intervals for each case study. In this way, the number of frames that have not been processed in the Kafka queue

throughout the frame flow will be determined and the stream processing efficiencies will be calculated at certain intervals. While this analysis is performed separately for each module, the flow efficiencies of the integrated system will also be calculated for cases where the modules are used integrated. In this context, the integrated system evaluation will be completed by examining the object-centric stream processing efficiency (OCSPE), event-centric stream processing efficiency (ECSPE) and integrated system stream processing efficiency (ISSPE) metrics. In order to measure the producing performance of the edge, the real-time recorded frames and the number of frames sent by the edge to the Kafka queue will be compared and the edge producing efficiency (EPE) metric will be used. The mathematical expression of these metrics is given below.

$$OCSPE = \frac{F_{consumed_object}}{F_{produced}} \quad (24)$$

$$ECSPE = \frac{F_{consumed_event}}{F_{produced}} \quad (25)$$

$$ISSPE = \frac{F_{consumed_event} + F_{consumed_object}}{2 \cdot F_{produced}} \quad (26)$$

$$EPE = \frac{F_{sent}}{F_{recorded}} \quad (27)$$

B. DATASET

After the problems that our approach will focus on, the points to be considered during development, and the clarification of the method, dataset research to be used in the evaluation of both sub-modules and in the training and evaluation of the event-centric summarization module was carried out. Within the scope of the dataset research, the datasets used in the studies in the first two sections and the characteristics of other datasets suitable for our study were determined. Since it will not be possible to obtain a dataset with the qualities that should be used in the evaluation of the integrated system in the separate development and evaluation of each module, it was concluded that using more than one dataset would be more effective. Before presenting the comparisons of the datasets we examined within the scope of the study, the dataset qualities required by each module were discussed separately. In addition, the dataset qualities required for the evaluation of the integrated system were also evaluated separately. First, the qualities that the dataset required for the development and evaluation of the object-centric summarization approach should have are given below;

- Since the object-centric video summarization module is implemented with a genetic algorithm approach using heat maps obtained with an optical flow-based approach, it does not contain any supervised approach.
- Therefore, it is only during the evaluation phase that the module will need the dataset.
- Since the main purpose of the module is to preserve the statistical properties related to the object in video analytics, it will be provided by comparing the results obtained by applying simple video analytics applications

such as traffic density analysis, crowd analysis, etc. on the summary video and the original video.

- In this context, in order to evaluate the relevant module, data sets are needed that include videos of people in indoor or outdoor areas and videos obtained by monitoring low and heavy vehicle traffic.

For the implementation and evaluation of the event-centric video summarization module, a dataset is needed, the attributes of which are presented as follows;

- A dataset consisting of videos of abnormal situations such as robbery, harassment, fight, arrest, vandalism, violence, use of firearms, assault, traffic accident, explosion, fire, theft is required for the training process of neural architectures to be developed within the scope of the relevant approach.
- It is necessary to mark the sections of the videos here, especially those where abnormal activities are performed.
- For the testing process, some videos need to be long enough to create a summary video by determining the abnormal activities in them with the method.

In this context, a comparison table of the data sets used for different video analytics applications in the first two sections and the data sets we encountered as a result of our online research with keywords such as “video traffic monitoring”, “crowd analysis”, “abnormal event detection” was prepared and is included in Table 2.

Considering Table 2 and the dataset requirements we mentioned earlier;

- Although the problems focused on in the datasets given with reference numbers [44], [45], [46], [47], [48] are video summarization approaches, their content is not compatible with video analytics applications in smart cities.
- Therefore, it is the right approach to use published data sets within the scope of video analytics.
- When considering data sets of this nature, the data sets given with reference numbers [49], [52], [53], [54] do not have the necessary qualities for the training and evaluation of the approaches to be developed.
- For the evaluation of the object-centric summarization module, traffic monitoring qualified videos were obtained from the dataset [64]. 254 videos in this dataset are 5-second segments of a long video taken from the same imaging device. We recombined these video segments to obtain a long video that needs to be summarized. (Case Study I and II)
- We considered using the dataset [59] to evaluate the performance of the object-centric summarization module in preserving the crowd analysis results, but it is a dataset consisting of only 2000 frames. Therefore, we performed our second evaluation by merging all the frames into a 20 fps video (Case Study III and IV).
- UCF-Crime [50], HWID-12 [60] and RLVS [61] data sets were used to train the model that will perform the classification process in event-centric summarization.

- At this point, each video in the RLVS and HWID-12 dataset already consists of videos with low duration. The videos in the UCF-Crime dataset also include long-term video sequences independent of the abnormal event.
- This is particularly not an ideal situation for training a model that will be used in a video summarization approach.
- To overcome this problem, 700 videos in the UCF-Crime dataset were subjected to a software-assisted manual video cropping process.
- Normal videos in the dataset were not used in the training process, only normal video sequences found in abnormal videos were included in the dataset. The length of all videos in the training dataset is 3 seconds.
- 2000 videos were taken from each dataset and a total of 2000 video sequences were manually extracted from the UCF-Crime dataset. The final dataset consists of 6000 videos.
- Of these 6000 videos, 1000 are reserved for testing. The remaining 5000 videos will be used for training the model (Case Study V).

By using the mentioned datasets together, it is aimed to create a training set with high quantitative and qualitative richness. The videos combined for testing the model have different environments, lighting and camera angles. In this way, it shows the adaptability of the developed approach for different surveillance applications.

VI. EXPERIMENTAL RESULTS

A. EXPERIMENTAL RESULTS OF OBJECT-CENTRIC VIDEO SUMMARIZATION MODULE

In order to evaluate the performance of the object-centric summarization module's approach using heat maps generated by object detection, the videos in the dataset [64] were first combined and compared with different genetic algorithm parameter configurations and fitness functions (Case Study I). The results of these comparisons are given in Table 3. For applying the GA in our approach we used the *deap* library. Hyperparameter ranges for GA were determined after many experiments. In configurations where hyperparameters had values far from the specified range, the genetic algorithm produced results with fitness values below 0.75. The configuration setting made for GA was continued until the fitness score exceeded 90%, which is the acceptable value, and the lowest possible generation and population numbers were obtained. The GA parameter configuration performed within the scope of this video will also be used in other Case Studies focused on object tracking. Table 4 shows the GCSP, LCSP, SAS, and OVSE scores of the summary videos obtained as a result of the experiments in Case Study I. YOLOv5s model was used for heat map generation and evaluation processes in Case Study I. Figure 10 shows the normal and abnormal keyframes obtained from the summary video. Figure 11 shows the heat maps of the summary video and original videos produced with the highest performance.

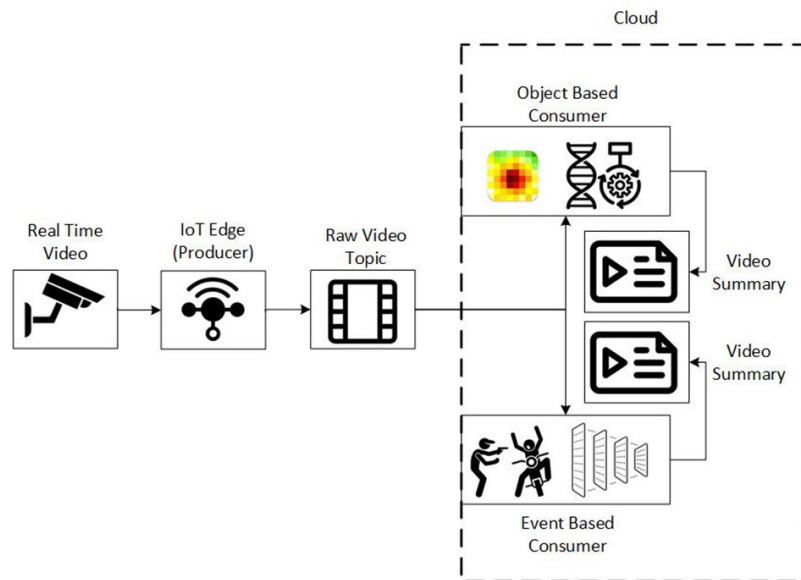


FIGURE 9. Integrated system with Apache Kafka.

TABLE 2. Table of dataset comparisons.

Reference Number	Dataset	Purpose	Quantitative information
[44]	VSUMM	Video summarization	50 Videos, 250 caption
[45]	OVP	Video Segmentation Video Summarization	3659 Video
[46]	TVSum	Video Summarization	50 Video, 1000 annotations
[47]	SumMe	Video Summarization	25 Video, 390 summaries
[48]	MST-VTT	Video Captioning	10000 Video
[49]	BDD100K	Driving Video	100000 Video
[50]	UCF-50	Action Recognition	6681 Video for 50 activities
[51]	UCF-Crime	Anomaly Detection	700 Videos for 14 abnormal events
[52]	COCO2017	Object Detection	165000 images
[53]	KITTI	Autonomous Driving	389 Video
[54]	Cityscapes	Video segmentation	20000 Images
[55]	VIRAT	Action recognition	12.5 hours video
[56]	CUHK	Crowd Counting	1 hour video
[57]	TAO	Object tracking	2907 video
[58]	ShanghaiTech Dataset	Single Image Crowd Counting	241677 Image
[59]	Mall Dataset	Crowd Counting	2000 Video Frames
[60]	HWID-12	Traffic accident videos	2780 videos
[61]	RLVS	Violence detection	2000 Videos
[62]	Urban Tracker	Object Tracking	4 videos
[63]	SCVD	Violence Detection	3632 videos
[64]	-	Traffic Monitoring	254 Video

In the scope of Case Study II, evaluations of the summary approach using optical flow-based heat maps were performed. The results of these comparisons are given in Table 5. Table 6 shows the GCSP, SAS, and OVSE scores of the summary videos obtained as a result of the experiments in Case Study II. In Case Study I, the YOLOv5s model was used in heat map production to perform the GCSP evaluation. Figure 12 shows the normal and abnormal keyframes obtained from the summary video. Figure 13 shows the optical flow heat maps of the summary video and the original videos produced with the highest success. Figure 14 shows

the heat maps of the summary video and the original video created with YOLOv5s.

For Case Studies III and IV, as mentioned before, the Mall dataset [59] was used. This dataset was used for the video summarization approach, which uses heat maps based on object detection results using the YOLOv5 model in Case Study III. Table 7 presents the comparative results of the genetic algorithm approach implemented for Case Study III. Table 8 presents the GCSP, SAS, and OVSE scores of the summary videos obtained as a result of the experiments in Case Study III. Figure 15 presents the normal and abnormal

keyframes obtained from the summary video. Figure 16 presents the heat maps of the summary video and the original videos produced with the highest performance.

In the scope of Case Study IV, evaluations of the summary approach using optical flow-based heat maps were performed. The results of these comparisons are given in Table 9. Table 10 shows the GCSP, SAS, and OVSE scores of the summary videos obtained as a result of the experiments in Case Study IV. Figure 17 shows the normal and abnormal keyframes obtained from the summary video. Figure 18 shows the optical flow heat maps of the summary video and the original videos produced with the highest success. Figure 19 shows the heat maps of the summary video and the original video created with YOLOv5s.

B. EXPERIMENTAL RESULTS OF EVENT-CENTRIC VIDEO SUMMARIZATION MODULE

For event-centric video summarization, 6000 videos of 3 seconds each obtained from the datasets mentioned in the datasets section were used for training, validation and testing processes. In neural architecture training, 16 frame CNN architectures were selected homogeneously from these 3-second videos and their features were extracted with recurrent or attention mechanisms to ensure that sequential relationships were learned. Keras and tensorflow library were used for the training and inference phases of the neural models. The maximum epoch number was determined as 50 in all training processes, but in cases where the validation loss did not converge, an early-stopping mechanism that performed this tracking with 5 epoch memory was used. Table 11 shows the accuracy and loss values of the training, validation and testing processes of the four different models trained. Table 12 provides quantitative information about the parameters of the models. Figure 20 shows the training and validation loss and accuracy graphs of the CNLSTM (I) neural architecture, and Figure 21 shows the training and validation loss and accuracy graphs of the PRECNLSTM (II) neural architecture. Figure 22 shows the training and validation loss and accuracy graphs of the CNTRANSFORMER (III) neural architecture, and Figure 23 shows the training and validation loss and accuracy graphs of the PRECNTRANSFORMER (IV) neural architecture. When the given results are examined, it is observed that the Attention mechanism approaches provide higher performance results. Therefore, the event-centric summarization module evaluation will be performed on the CNTRANSFORMER and PRECN-TRANSFORMER models.

After training the models, 50 abnormally labeled and 250 normally labeled videos were selected from 1000 3-second test videos prepared for the evaluation of the event-centric video summarization approach and merged in a random order. The normal and abnormal status of each part of this merged test video was expressed with a 300-element binary encoded vector. Figure 24 shows 16 frames selected homogeneously from one 3-second normal and one 3-second

abnormal part of the video to be classified. Within the scope of Case Study V, the event-centric video summarization module was implemented to summarize this video and to produce the binary encoded label vector predicted by both models. Table 13 shows the SACC, SLOSS and EVSE metric evaluations of the binary encoded vector obtained by the models. Figure 25 shows one keyframe of the abnormal video part given in Figure 24 obtained with both neural architectures.

C. EXPERIMENTAL RESULTS OF INTEGRATED SYSTEM

In order to evaluate the integrated system, Kafka was first installed on a Raspberry Pi 4 development board and the Producer module that produces frames on this device was developed. At this point, the object-centric summarization module of the system, as is known, considered two different scenarios. The first of these is the scenario where an object detection process is already performed on the edge in the smart city system to which the system is integrated. Therefore, for this scenario, in addition to video frames, detection data made with the YOLOv5s model is also produced via Raspberry Pi 4. Therefore, in the tests to be carried out to perform the integrated system tests, it is necessary to evaluate the system under two different Case Studies by considering these two scenarios. At this point, in both Case Studies, a 10-minute indoor CCTV video including a theft sequence will be taken from the UCF-Crime [50] dataset and produced via Raspberry Pi. In Case Study VI, these video frames and the results of the detection process made with YOLOv5 on these frames will be given via Raspberry Pi 4. Figure 26 shows a graph showing the number of frames taken from a 30 fps video in 60-second periods, the number of data produced in frame and detection Kafka topics, the number of frames processed by the object-centric consumer, and the number of frames processed by the event-centric module. Table 14 shows the OCSPE, ECSPE, ISSPE, and EPE metric evaluations of the system in 1-minute periods in Case Study VI. Figure 27 shows the heat maps of the original video and the object-centric summary video. Table 15 shows the object-centric video summarization performance metrics of Case Study VI. Table 16 shows the event-centric video summarization performance metrics of Case Study VI.

In Case Study VII, the same video was re-implemented for the case where the object-based summarization module was implemented with the optical flow approach. Figure 28 shows a graph showing the frame numbers taken from a 30 fps video in 60-second periods, the number of data produced in frame and detection Kafka topics, the number of frames processed by the object-centric consumer, and the number of frames processed by the event-centric module. Table 17 shows the OCSPE, ECSPE, ISSPE, and EPE metric evaluations of the system in 1-minute periods in Case Study VII. Figure 29 shows the heat maps of the original video and the object-centric summary video. Table 18 shows the object-centric video summarization performance metrics of Case Study VI.



FIGURE 10. Keyframes selected from summary video produced in Case Study I. (a:Normal, b:Abnormal).

TABLE 3. Experimental results of Case Study I.

Exp. no	Population	Generation	Mutation Ratio	Stabilization Generation	Original Frame Count	Target Frame Count	Fitness
1	50	100	0.05	81	13317	50	0.8390
2	30	200	0.15	95	13317	75	0.8455
3	30	200	0.01	168	13317	100	0.863
4	20	300	0.02	267	13317	100	0.861
5	50	300	0.03	199	13317	100	0.862
6	50	300	0.02	115	13317	200	0.911
7	30	200	0.02	121	13317	200	0.917

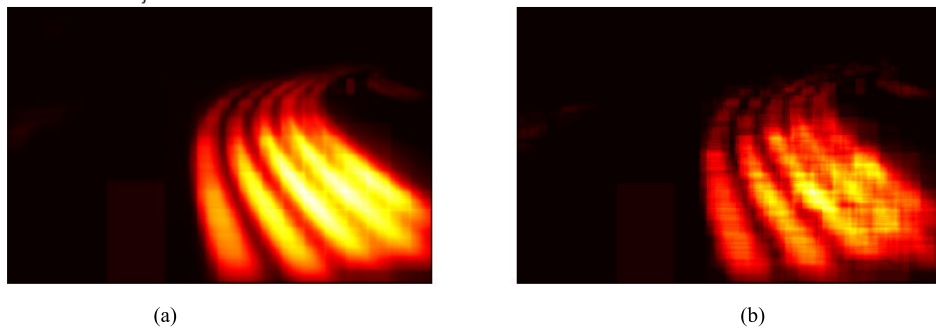


FIGURE 11. Heatmaps of original video (a) and best summary (b) produced by GA in Case Study I.



FIGURE 12. Keyframes selected from summary video produced in Case Study II. (a:Normal, b:Abnormal).

D. COMPARATIVE RESULTS

Although the modules we present have a different summary purpose than other studies in the literature, they should be compared with video summarization approaches in the literature in order to objectively evaluate their performance. In this context, we evaluated our two summarization approaches with the literature studies in the Table 19. In this evaluation, we used GCSP performance of object-centric summarization module and SACC performance of event-centric summarization module as summary accuracy. For video shortening

performance comparison we used VSE metric. For computational comparisons we used OCSPE and ECSPE as SPE metric. Also, F1-Score, mAP and any other accuracy metrics given in the literature study generalized as summary accuracy in the Table 19.

E. POWER CONSUMPTION ANALYSIS

To evaluate the energy footprint of different architectural deployments within our video summarization framework, we conducted simulated power measurements using the

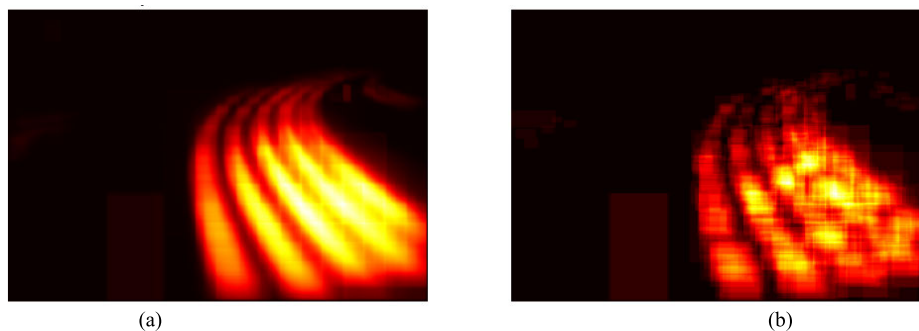


FIGURE 13. Heatmaps of original video (a) and best summary (b) produced by GA in Case Study II.

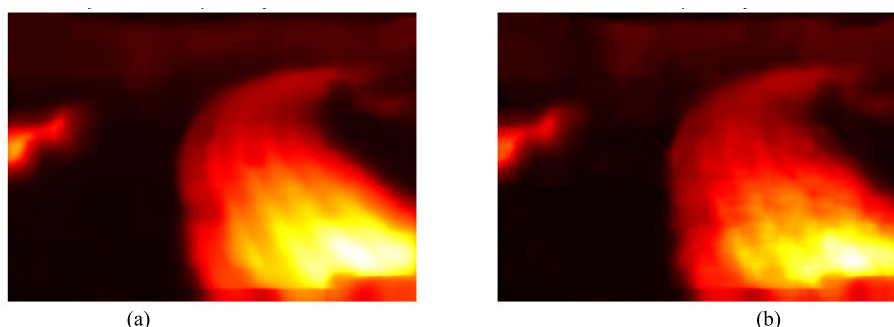


FIGURE 14. Object-detection heatmaps of original video (a) and best summary produced by GA (b) in Case Study II.

TABLE 4. GCSP, SAS and OVSE results of Case Study I.

Exp. no	GCSP	SAS	OVSE
1	0.8390	0.8132	99.24%
2	0.8455	0.8535	99.24%
3	0.863	0.8413	99.24%
4	0.861	0.8516	99.24%
5	0.862	0.8732	99.24%
6	0.911	0.9066	98.49%
7	0.917	0.9291	98.49%

TABLE 5. Experimental results of Case Study II.

	Pop	Gen	S.G	O.F.C	T.F.C	Fitness
1	30	100	37	13317	25	0.829
2	50	200	46	13317	25	0.846
3	50	200	35	13317	50	0.955
4	30	50	39	13317	50	0.960

TABLE 6. GCSP, SAS and OVSE results of Case Study I.

Exp. no	GCSP	SAS	OVSE
1	0.7016	0.8316	99.81%
2	0.7127	0.8545	99.81%
3	0.7835	0.8930	99.62%
4	0.8032	0.9135	99.62%

powerstat tool. The simulations focused on four distinct scenarios where key processing components—including frame capture, Kafka-based transmission, object detection, and

TABLE 7. Experimental results of Case Study II.

	Pop	Gen	S.G	O.F.C	T.F.C	Fitness
1	30	200	114	2000	200	0.974

TABLE 8. GCSP, SAS and OVSE results of Case Study II.

Exp. no	GCSP	SAS	VSE
1	0.974	0.9973	90%

TABLE 9. Experimental results of Case Study II.

	Pop	Gen	S.G	O.F.C	T.F.C	Fitness
1	30	200	114	2000	200	0.974

TABLE 10. GCSP, SAS and OVSE results of Case Study II.

Exp. no	GCSP	SAS	VSE
1	0.9194	0.9612	97.5%

TABLE 11. Performances of neural models trained in this study.

	Training		Validation		Test
Model	Acc	Loss	Acc	Loss	Acc
I	68.65%	59.34%	70.44%	58.75%	70.10%
II	71.10%	53.94%	70.66%	55.67%	71.22%
III	96.78%	6.44%	92.16%	30.03%	93.92%
IV	93.12%	17.95%	88.69%	31.04%	90.19%

optical flow analysis—were selectively assigned to either the edge (Raspberry Pi 4) or the cloud (Intel i5 with 6GB GPU).

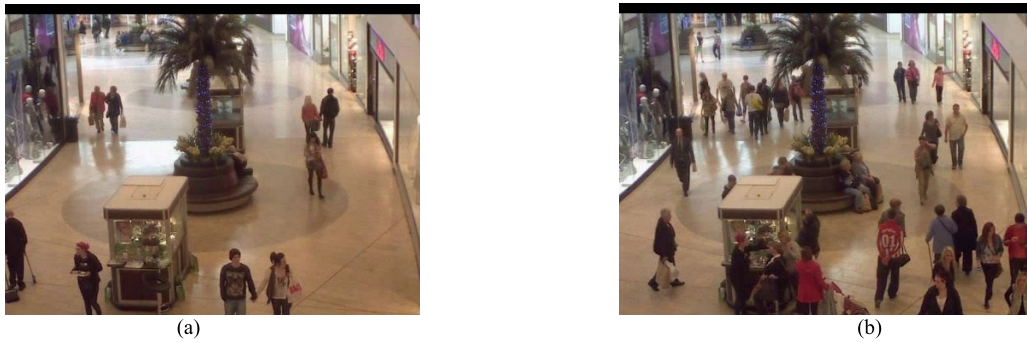


FIGURE 15. Keyframes selected from summary video produced in Case Study III. (a:Normal, b:Abnormal).

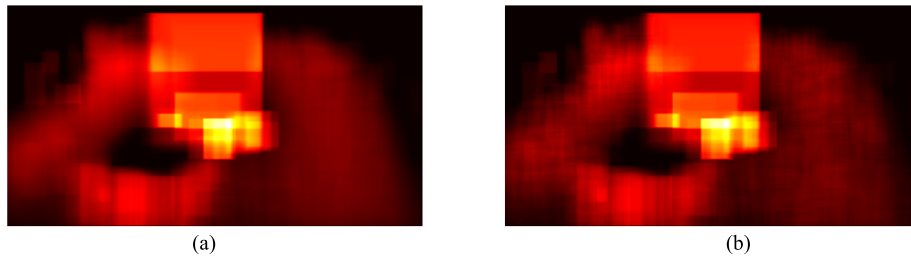


FIGURE 16. Object detection Heatmaps of original video (a) and best summary (b) produced by GA in Case Study III.

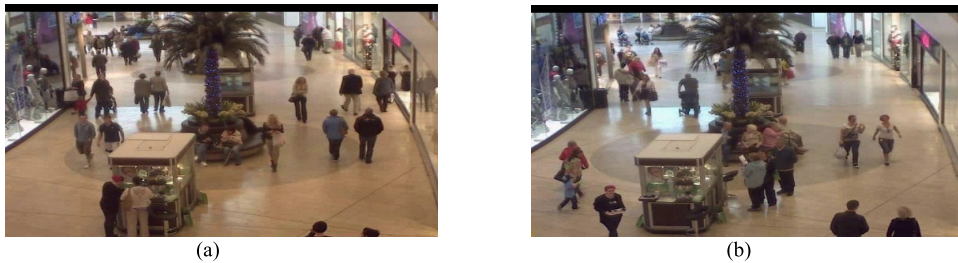


FIGURE 17. Keyframes selected from summary video produced in Case Study IV. (a:Normal, b:Abnormal).

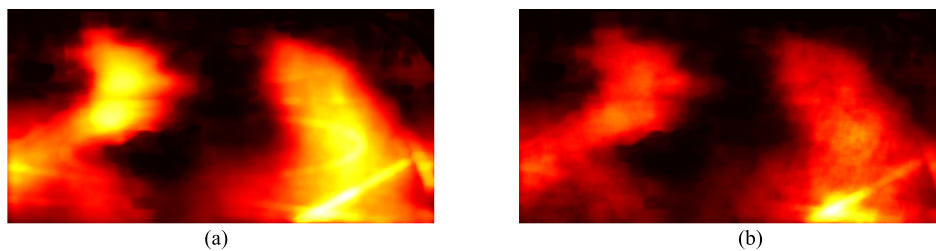


FIGURE 18. Optical flow heatmaps of original video (a) and best summary produced by GA (b) in Case Study IV.

Table 20 shows simulation results of energy consumption for different cases. In all configurations, summarization modules (object-centric and event-centric) were executed in the cloud to ensure consistent comparison. Energy consumption values are reported in Watt-hours (Wh) and were computed over a standardized 10-minute video stream. Baseline (idle) energy

values were recorded on both devices as a reference, and the energy consumed by each task was calculated by subtracting this baseline from the total measured power usage. For example, in the scenario where object detection is performed on the edge, the Raspberry Pi 4 consumed approximately 2.17 Wh, while the cloud required 1.49 Wh, resulting in a

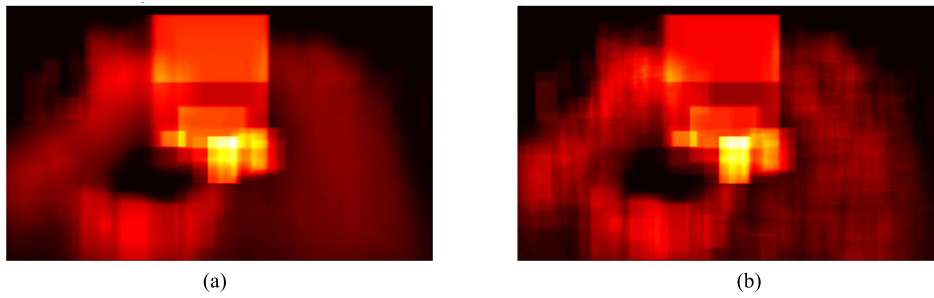


FIGURE 19. Object-detection heatmaps of original video (a) and best summary produced by GA (b) in Case Study IV.



FIGURE 20. Loss and accuracy graphs of CNN-LSTM.



FIGURE 21. Loss and accuracy graphs of PRECNN-LSTM.

total of 3.66 Wh. When only Kafka and optical flow analysis were executed on the edge, the edge energy consumption was 1.72 Wh, and the cloud consumed 1.47 Wh, totaling 3.24 Wh. In the scenario where object detection was moved to the cloud, the edge consumption decreased to 1.24 Wh, while cloud-side processing increased to 1.69 Wh, yielding a total of 2.93 Wh. Lastly, when both optical flow and object detection were executed in the cloud, the edge consumed 1.21 Wh and the cloud 1.54 Wh, for a total of 2.75 Wh. Across all scenarios, Kafka transmission incurred a measurable and consistent energy overhead on the edge. These results emphasize the trade-off between local computation and cloud offloading

in terms of power consumption, providing actionable insights for designing energy-aware summarization systems aligned with Green AI principles.

VII. DISCUSSION OF EXPERIMENTAL RESULTS

When the results of Case Study I are examined, it is seen that the genetic algorithm achieved the highest result with the parameters used in experiment number 7 (Table 3). Therefore, in other object-centric summarization applications that use heat maps created based on object detection, the GA parameter configurations found in experiment number 7 were also used. When Table 4 is examined, it is clearly seen that

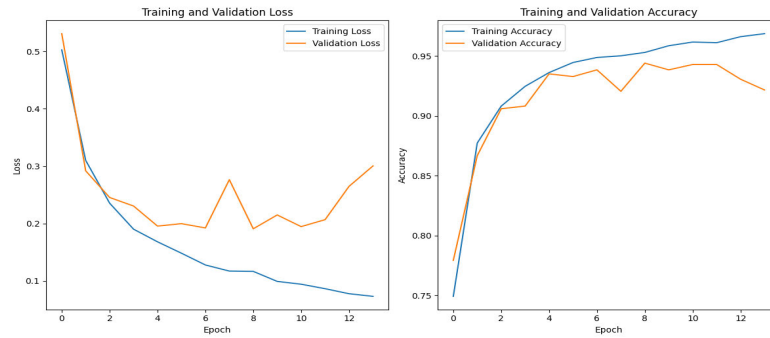


FIGURE 22. Loss and accuracy graphs of CNN-TRANSFORMER.

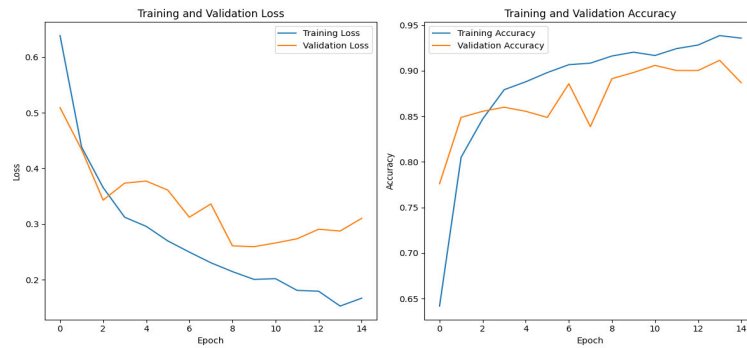


FIGURE 23. Loss and accuracy graphs of PRECNN-TRANSFORMER.

TABLE 12. Quantitative informations of neural network parameters.

Model	Total Parameter Count	Trainable Parameter Count	Total Parameter Size	Trainable Parameter Size
CNNLSTM	20762690	20762690	79.20 MB	79.20 MB
PRECNLSTM	20762690	783306	79.20 MB	2.82 MB
CNNTRANSFORMER	20928450	20928450	79.84 MB	79.84 MB
PRECNNTRANSFORMER	20928450	904066	79.84 MB	3.45 MB



FIGURE 24. Frames taken from normal and abnormal video batches (a:Normal, b:Abnormal).

the increase in structural similarities of heat maps and other performance metrics show a directly proportional relationship. This also reveals the importance of implementing the genetic algorithm approach within the scope of heat maps. In Case Study II, it is seen that the optical flow-based object-

centric summarization process was performed and the most suitable GA configuration was performed with the parameters in experiment number 4 (Table 5). When the performance metrics given in Table 6 are examined, it is seen that the optical flow-based object-centric summarization approach



FIGURE 25. Detected keyframes by CNN-TRANSFORMER (a) and PRECNN-TRANSFORMER (b).

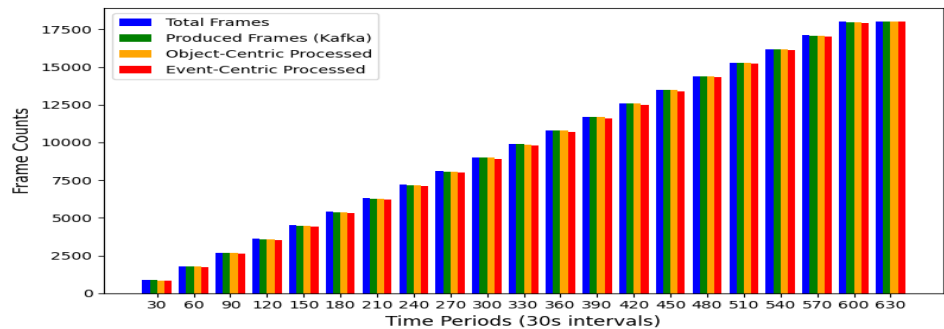


FIGURE 26. Frame stream progress graph.

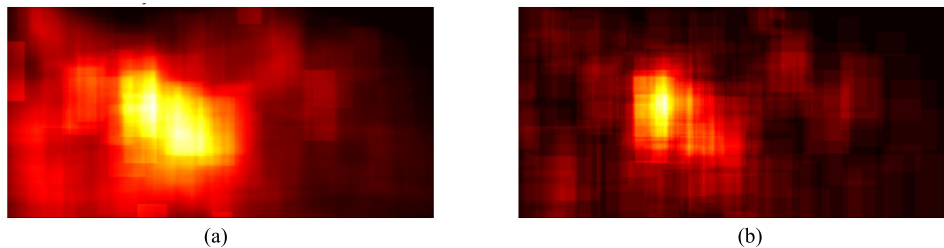


FIGURE 27. Object-detection heatmaps of original video (a) and best summary produced by GA (b) in Case Study VI.

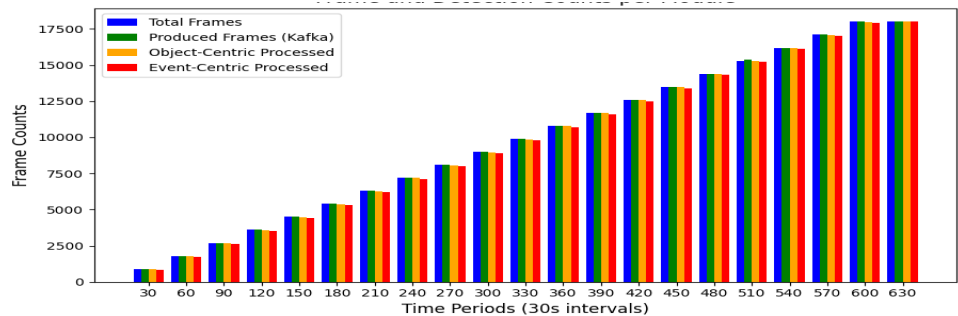


FIGURE 28. Frame stream progress in Case Study VII.

achieves higher performance results than the object detection-based approach. When the results of Case Studies III and IV are examined, it is seen that the object-centric summarization approach has a very high performance in terms of preserving

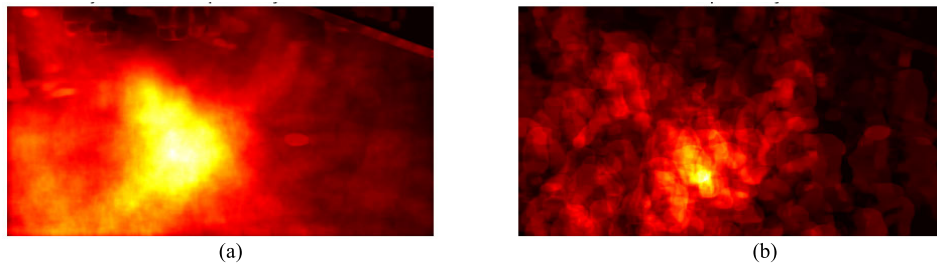


FIGURE 29. Optical flow heatmaps of original video (a) and best summary produced by GA (b) in Case Study VII.

TABLE 13. SACC, SLOSS, EVSE and VSE results of the models.

Models	SACC	SLOSS	EVSE	VSE
I	0.7763	0.6542	0.7234	71.2%
II	0.8112	0.5446	0.7714	74.6%
III	0.9899	0.0033	1.0000	80%
IV	0.9375	0.0500	0.9250	70.4%

TABLE 14. Integrated system performance of Case Study VI.

Time Period	OCSPE	ECSPE	ISSPE	EPE
0-60s	0.9888	0.9722	0.9805	0.9736
60-120s	0.9942	0.9714	0.9828	0.9729
120-180s	0.9945	0.9780	0.9862	0.9791
180-240s	0.9943	0.9719	0.9831	0.9786
240-300s	0.9944	0.9750	0.9847	0.9789
300-360s	0.9944	0.9749	0.9846	0.9814
360-420s	0.9944	0.9779	0.9861	0.9812
420-480s	0.9944	0.9777	0.9887	0.9838
480-540s	0.9971	0.9804	0.9888	0.9863
540-600s	0.9972	0.9804	0.9888	0.9836

TABLE 15. GCSP, SAS and OVSE values of Case Study VI.

Exp. no	GCSP	SAS	VSE
1	0.8442	0.9380	93.33%

TABLE 16. SACC, SLOSS and EVSE values of Case Study VII.

Model	SACC	SLOSS	EVSE	VSE
CNN-TRANSFORMER	0.8571	0.0050	75.0%	98.5%

TABLE 17. SACC, SLOSS and EVSE values of Case Study VII.

Time Period	OCSPE	ECSPE	ISSPE	EPE
0-60s	0.9744	0.9000	0.9372	0.9988
60-120s	0.9883	0.9500	0.9691	0.9994
120-180s	0.9825	0.9666	0.9746	0.9996
180-240s	0.9963	0.9750	0.9856	0.9997
240-300s	0.9911	0.9800	0.9855	0.9997
300-360s	0.9946	0.9833	0.9889	0.9998
360-420s	0.9955	0.9857	0.9906	0.9996
420-480s	0.9980	0.9888	0.9927	0.9997
480-540s	0.9959	0.9900	0.9924	0.9998
540-600s	0.9966	0.9900	0.9940	0.9986

statistical information in scenarios with intense human movements (Table 7, 8, 9, 10).

TABLE 18. GCSP, SAS and OVSE values of Case Study VII.

Exp. no	GCSP	SAS	VSE
1	0.856	0.9267	99.72%

TABLE 19. Comparative results.

Study	Approach	Case Quality	SACC	VSE	SPE
[3]	Feature-centric	Best	83.25%	-	-
[3]	Feature-centric	Worst	80.0%	-	-
[4]	Event-centric	Best	60.7%	-	-
[4]	Event-centric	Worst	44.1%	-	-
[5]	Feature-centric	Overall	50.59%	97.38%	-
[6]	Event-centric	Overall	78.78%	-	-
[7]	Object-centric	Best	64.1%	98.61%	-
[7]	Object-centric	Worst	28.8%	79.00%	-
[8] ⁺	Object-centric	Best	100%	96.4%	-
[8] ⁺	Object-centric	Worst	88.70%	88.8%	-
[9]	Object-centric	Best	91.27%	-	-
[9]	Object-centric	Worst	86.33%	-	-
[10]	Event-centric	Overall	46.99%	-	-
[17]	Event-centric	Best	76.0%	-	99.5%
[17]	Event-centric	Worst	46.0%	-	83.4%
Ours	Object-centric	Best	91.94%	99.7%	99.6%
Ours	Object-centric	Worst	85.6%	93.7%	97.4%
Ours	Event-centric	Best	98.99%	98.5%	99.0%
Ours	Event-centric	Worst	85.71%	80%	96.6%

TABLE 20. Simulation results for energy consumption under different processing configuration.

Kafka Producer	YOLO	Optical Flow	Summarization	Edge (Wh)	Cloud (Wh)	Total (Wh)
Edge	Edge	-	Cloud	2.17	1.49	3.66
Edge	-	Edge	Cloud	1.72	1.47	3.24
Edge	Cloud	-	Cloud	1.24	1.69	2.93
Edge	-	Cloud	Cloud	1.21	1.54	2.75

When the neural architectures trained for the event-centric summarization module are examined, it is seen that transformer layer neural architectures have a clear superiority over the others (Table 11 and Table 13). Additionally, when Tables 11 and 13 are examined, it is seen that the summarization operations performed with non-pretrained models have higher performance.

When the event-centric summarization metrics of Case Study V are examined, it is seen that results that are well above the targeted 95% summary performance are obtained (Table 13). Case Studies I, II, III, IV and V were basically

carried out for the purpose of evaluating the sub-components of the system separately. If we were to interpret all the cases here in one sentence, it was seen that the developed approaches had very high performances both in terms of summarization purpose and time reduction. The performance evaluation of the integrated system was carried out in Case Study VI.

In this approach, the image frames generated at the edge are first subjected to a YOLOv5s-based object detection process. This is especially evident when the scenario in Case Study VII is compared with the EPE metric, and it is seen that the object detection process causes a decrease in the frame production performance of the edge (Table 14, 18).

Apart from this, it is observed that the OCSPE, which evaluates the object-centric summarization approach, decreases in Case Study VII, albeit by a very small amount, due to the optical flow processing costs. This small amount reveals how logical the optical flow approach is in terms of processing costs. As is known, the event-centric summarization component continues its operations on 3-second video batches. Therefore, it is not possible to perform its operations before all the frames belonging to these 3-second video batches are produced. Therefore, in Figures 26 and 28, the number of frames processed by the Consumer, which performs the operations belonging to this module, is lower than the other components. However, since the number of frames processed by this module always remains a constant distance from the number of frames produced, it does not cause any accumulation in the Kafka queue. For the testing of the integrated system, the video used in Case Study VI and Case Study VII was selected to push the limits of both modules. Because, for object-centric video summarization, it is difficult to summarize videos with low object density with both genetic algorithm-based methods. Therefore, it is seen that the object-centric summarization module falls behind the performance observed in Case Studies I-IV in the current parameter configuration. As a precaution against this performance decrease, a mechanism that dynamically adjusts the population and generation number for cases where the fitness value remains below 90% can be integrated into the genetic algorithm flow. In addition, the sequences considered as abnormal events in the video are the picking up of a fallen object belonging to someone else by another person. Despite this, the developed CNN-TRANSFORMER model could not detect the anomalies in some sequences, but it achieved 85.0% summary performance. Again, the video shortening performances of both components are quite high.

Finally, when Table 19 is examined, the superiority of our approach over existing summarization approaches is seen. Since there are not many summarization approaches developed directly for surveillance applications in the current literature studies, event or object-centric summarization approaches developed for different smart city applications are also included in the comparison.

VIII. CONCLUSION

In this study, a multi-objective video summarization approach is proposed for summarizing surveillance videos used in smart city systems. The proposed method has two main objectives: to preserve object-based statistical information in videos and abnormal movements encountered in areas such as health, security, and transportation. For this purpose, two modules, one object-centric and the other event-centric, were developed. The developed object-centric module was implemented with a GA-based method that works with a fitness function focused on the structural similarity of heat maps that have a high ability to cover different statistical object features. For the development of the event-centric module, multiple neural architecture trainings configured to classify video sequences as abnormal and normal were performed and the results were compared. Since the CNN-Transformer neural architecture has the highest performance among these architectures, the classification process was performed with this model. These two sub-modules were combined with Apache Kafka to obtain an integrated system. Each module and integrated system were evaluated separately from different perspectives and these evaluation results were given in detail. When these given results are examined, it is seen that the object-centric module preserves statistical properties with a success rate of over 90%, while the event-centric module summarizes abnormal events with a success rate of over 95%. Also, the stream processing efficiency of the integrated system is quite high ($>95\%$). Finally, the different qualities of the proposed method are compared with other approaches in the literature. The purpose of this comparison is to show that this original method we developed is superior to other studies in the literature with its performance. As a result of the comparison, the superiority of the proposed method over the studies in the literature in terms of summary performance, video shortening ratio and stream processing efficiency is clearly seen. The proposed method will enable smart city systems to be more sustainable and have lower processing costs with its high performance, original and multi-purpose summary approach and easy integration into smart city systems of different qualities. In future work, it is planned that the system architecture will be extended to support more flexible and adaptive task distribution strategies between edge and cloud environments, in accordance with varying computational and network constraints. In addition, efforts will be directed towards enhancing the system's compatibility with emerging IoT infrastructures, with particular attention to scalability, latency tolerance, and interoperability with heterogeneous sensor networks. In this way, the applicability of the proposed approach in large-scale and dynamic smart city settings is expected to be further improved.

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